# The coppe document class Version 3.5.1

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**Abstract**

In this work, it is described the coppe document class as well as other files distributed by the CoppeTEX project. This class is suitable for writing academic dissertations, thesis and qualifying exams according to the formatting rules of the Alberto Luiz Coimbra Institute for Graduate Studies and Research in Engineering. The minimalist set of macro commands allows its users to concentrate most of their efforts on text composition rather than on the document layout.

## 1 Introduction

Writing documents in LaTeX may be a laborious task when the authors have to prepare their manuscripts rigorously respecting formatting rules imposed by publishers. Regardless of difficulty, a lot of thesis presented to the Coordination of Graduate Studies and Research in Engineering of the Federal University of Rio de Janeiro (COPPE/UFRJ) is typeseted in LaTeX. This demand motivated the creation of the CoppeTEX project, which tries to facilitate and encourage the use of LaTeX within the COPPE/UFRJ scope.

The coppe document class is the main product of CoppeTEX. It was designed to be clear and succinct. It enables the creation of dissertations, qualifying exams and thesis in a simple and automatic way. The main goal of the coppe class is to maintain authors strictly focused on text composition without worrying about margins sizes, line spacing, paper size, vertical and horizontal alignment, etc. The CoppeTEX project comprehends also BibTEX and MakeIndex style files for creating lists of references, symbols and abbreviations. Although there aren't official guidelines to write qualifying exams, we provide this option just for convenience, as this exam is a requisite to obtain the DSc degree and, for some of the programs, the MSc degree.

In which follows, it is described the user interface of the coppe class. Some details about using the style files cited above are also given. We use the term *thesis* to generally refer to dissertation, qualifying exam, and thesis itself.

## 2 License

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To see the full text of the GNU GPL license, go to the `COPYING` file attached to this package.

## 3 Support

We maintain a mailing list where users can send questions, comments, and bugs to. More details can be found here.

However, as the project maintenance was transferred to a new repository, bug reports, as well as new feature requests, should be directed to `https://github.com/COPPE-UFRJ/CoppeTeX`.

## 4 User interface

`\frontmatter` `\mainmatter` `\backmatter` A thesis to be approved by the Academic Registry at COPPE/UFRJ must contain three-parts: *front*, *main* and *back* matters [?]. Each one of these parts is started by calling its corresponding macro `\frontmatter`, `\mainmatter` or `\backmatter`. The front matter of a thesis consists of front cover and face, cataloging page, dedication, acknowledgments, abstracts, table of contents, and lists of tables, algorithms, symbols and abbreviations. The main matter is just composed by chapters, while the back matter usually consists of bibliographic references, appendices and index.

You must invoke the `\frontmatter` macro immediately after the `\maketitle` one. The `\mainmatter` command comes right before the first chapter, and `\backmatter` must be typed before the list of references.

### Front cover

This element was recently introduced by the Academic Registry. It is automatically constructed by the `\maketitle` command.

### Front face

The front face is unnumbered. There, it is not allowed to use hyphenation [?]. It is constructed by calling `\maketitle`. Next, it is described the commands used to enter the information required to create it.

`\author` The `\author` command was redefined. Here, it takes two arguments: the author's first names and surname, e.g., `\author{First Names}{Surname}`. The words should be typed with only first letters in uppercase.

`\title` `\foreigntitle` The macros `\title` and `\foreigntitle` are used to enter the titles of your monograph in the current and foreign languages. The default languages are Brazilian Portuguese and English. The `babel` package is automatically loaded by `coppe.cls`, so you do not need to load it again. The Brazilian Portuguese is the main language and the English is only required for the foreign abstract.

`\advisor` `\examiner` Every COPPE student is coordinated by at least one advisor. M.Sc. and D.Sc. students can have at most 2 and 3 advisors, respectively. Their names must be provided by issuing the command `\advisor` as below:

```
\advisor{Title}{Advisor's Name}{Surname}{Degree}
```

```
\advisor{Title}{Second Advisor's Name}{Surname}{Degree}
\advisor{Title}{Third Advisor's Name}{Surname}{Degree}
```

The advisors are not necessarily members of the thesis examination board. Thus, it is required to enter the names of all examiners using the `\examiner` macro. The examiners' names are entered differently:

```
\examiner{Title}{First Examiner's Name Surname}{Degree}
\examiner{Title}{Second Examiner's Name Surname}{Degree}
...
\examiner{Title}{N-th Examiner's Name Surname}{Degree}
```

Remember that all names must be given before calling `\maketitle`.

`\department` The Alberto Luiz Coimbra institute is divided into 12 academic units: Biomedical Engineering (PEB), Civil Engineering (PEC), Electrical Engineering (PEE), Mechanical Engineering (PEM), Metallurgical and Materials Science Engineering (PEMM), Nuclear Engineering (PEN), Ocean Engineering (PENO), Energy Planning (PPE), Production Engineering (PEP), Chemical Engineering (PEQ), Systems Engineering and Computer Science (PESC), and Transportation Engineering (PET). You must specify your department using one of the above abbreviations, e.g., `\department{PEC}`.

`\date` This macro is used to set the month and year of defense. This information is required to create the front face, cataloging details page and abstracts. For example, October 2007 should be entered as `\date{10}{2007}`.

`\keyword` The keywords should describe the concentration areas of your work. You must provide them as follows:

```
\keyword{First Keyword}
\keyword{Second Keyword}
...
\keyword{N-th Keyword}
```

Usually, six words are enough.

## Cataloging details

This page contains cataloging information useful for librarians. Fortunately, it is automatically generated from the data you entered at the time you call `\maketitle`. It is not needed in qualifying exams, though.

## Dedication (optional)

`\dedication` This macro was added for convenience. The input text is placed at the right bottom of a blank page. It is emphasized and in normal size.

## Abstracts

`abstract` (*env.*)
`foreignabstract` (*env.*)

As stated by the Academic Registry [?], abstracts must be in one page each, with at most 250 words. We recommended that they should be only one paragraph long. They must be defined inside the environments `abstract` and `foreignabstract`.

## Lists of symbols and abbreviations (optional)

`\abbrev` `\symbl`

The lists of symbols and abbreviations are optional, although highly recommended. It is a good practice to define a symbol/abbreviation in its first occurrence in the text. To define a symbol use `\symbl[alphabetic symbol]{Symbol}{Symbol Definition}`, and for abbreviations `\abbrev[alphabetic symbol]{Abbreviation}{Abbreviation Definition}`. These commands are called *dummy*, since they don't output anything at the place they are executed, just an entry in the correspondent list.

`\makeloabbreviations` `\makelosymbols` `\printloabbreviations` `\printlosymbols`

These lists are lexicographically sorted by using the MakeIndex program, which is part of any LaTeX implementation. For `\symbl`, if the optional parameter is provided, it will used as sort key. This was later, in 2024, implemented also for `\abbrev`, otherwise `Symbol`, or `Abreviation` will be used as sort key, what can result in an undesirable order if it contains LaTeXcommands, mathematical symbols, or mix of uppercase and lowercase. MakeIndex needs two commands to create a final sorted list: one which generates a list of entries and the other that indicates the position where the list will be printed out. To generate the lists of symbols and abbreviations, the `coppe` class provides the commands `\makeloabbreviations` and `\makelosymbols`, respectively. They must be called in the document preamble. The commands `\printlosymbols` and `\printloabbreviations` have to be invoked at the point where you want these lists appear, e.g., following the list of tables as showed in the example. Once you call `latex`, it will be created two files with extensions `abx` and `syx`, which contain MakeIndex input data. They must be processed with `makeindex` in order to get the lists correctly produced, redirecting the output to files with extension `lab` and `los` respectively:

```
makeindex -s coppe.ist -o example.lab example.abx
makeindex -s coppe.ist -o example.los example.syx
```

Note the `-s` option for specifying the style `coppe.ist`. Now, rerun `latex` twice to get the references solved and you are done.

## References

CoppeTeX produces citations and the reference list with `biblatex` and the `biber` backend, through its own bibliography style implementing the post-2020 ABNT NBR 6023 rules. The class loads this style automatically: there is no `\bibliographystyle` to set and no BibTeX `.bst` file to choose (the former `coppe-plain` and `coppe-unsrt` BibTeX styles have been retired).

By default, citations follow the author–date style, e.g., "Chartier (2002)". Passing the `numbers` class option switches to the numeric style, in which citations are bracketed numbers and the reference list is numbered in order of citation (unsorted).

Declare your bibliography database in the preamble with `\addbibresource` and print the list where it should appear with `\printbibliography`:

```
\addbibresource{example.bib}   % in the preamble
...
\printbibliography             % where the list goes
```

Entry types and fields may be written in English (`@book`, `title`, `author`) or with their unaccented Portuguese synonyms (`@livro`, `titulo`, `autor`).

Run in sequence LaTeX, biber, and twice again LaTeX to resolve the citations and references. The style is natbib compatible (via the biblatex natbib=true option), so you may freely issue \citet and \citep, as well as the other natbib commands.

### Appendix and Annex (Optional)

\appendix
\annex

Appendices and annexes are optional chapters that are part of the back matter. The \appendix command is a standard LaTeX command used before all appendices. CoppeTeX introduces the \annex command, which should be used only in the back matter (i.e., after the \backmatter command) and only after all existing \appendix chapters. This restriction is due to implementation constraints.

Therefore, the order for backmatter is:

```
\backmatter
\printbibliography
\appendix
\chapter{An Appendix}
\chapter{Another Appendix}
\annex
\chapter{First Annex}
\chapter{Second Annex}
```

Annex was introduced in v3.5

## 5 Class options

There are some options users can specify in order to customize the appearance of the output produced by the coppe class. These options can be passed to coppe as follows: \documentclass[option1, option2]{coppe}. In which follows, we give a brief description of all supported options.

**dsc, msc, dscexam, mscexam** The coppe class is able to produce thesis, dissertations, and qualifying exams, which are enabled by the dsc, msc, mscexam, and dscexam options, respectively.

**doublespacing** The default line spacing is one-and-a-half. For enabling double spacing between lines, use the doublespacing option.

**numbers** The default citation style is the author–date scheme. For numbered citations, specify the numbers option to the coppe class; the reference list is then numbered in order of citation (unsorted). The matching biblatex style is selected automatically in either case.

**english** CoppeTeX uses Babel. The default language is Portuguese (actually brazilian), with English being the second language. If option english is used, English becomes the main language and Portuguese the secondary. Look at the Babel package to switch between languages.

### 5.1 Changing document identification

\freeconfig The user could *optionally* use the command `freeconfig` to modify the parameters that print the document identification. The command `freeconfig` needs all those paramerters, which are degree initials, degree name, title, foreign title, local doctype, and foreign doctype as in the following example:

```
\freeconfig{Dr.}{Philosophiae Doctor}{PhD}{Doutor}{Dissertation}{Tese}
```

## 6 Quick, useful tips

**Pictures.** The default picture format of LaTeX is the Encapsulated PostScript (EPS). If you use pdfLaTeX, the default format becomes the PDF, but you can equally load PNG files. For such, you must enter the name of your image file without extension, e.g., `\includegraphics{filename}`, and `pdflatex` will firstly look for a file called `filename.pdf` and after for file `filename.png`. For producing high quality pictures with embedded fonts we recommend the Ipe drawing software available here.

**Fonts.** The default font in LaTeX is the Computer Modern. If you would like to try its enhanced version, consider using the `lmodern` package. To use Times, it is recommended to load the package `mathptmx`, rather than the deprecated `times`. There is also an enhanced Times version available with the `tgtermes` package. You can still use the Arial font face with the package `uarial`.

**Hyperref.** When working with PDF's, there is the possibility to add extra information to the file as the author's name, document title, subject, keywords, etc. This is easily done with the `hyperref` package. It is also useful to enable hyperlinks. Fortunately, the `coppe` class will do this automatically if `hyperref` is loaded.

**Printing.** To get your work correctly printed, you must ensure that any page scaling option (e.g., fit or shrink to printable area) isn't enabled. This kind of option often comes in print dialogs of document visualization softwares.

longquote **Quotation** To quote text larger than three lines, according to ABNT, you must increase the left margin to 4 cm, do not use quotation marks, and use a smaller font. The `coppe` class provides the `longquote` environment to easily make these adjustments.

## 7 A simple example

```
⟨*example⟩
\documentclass[dsc]{coppe}

\usepackage{booktabs}% tabelas mais bonitas
\usepackage{rotating}% rodando coisas, como tabelas
\usepackage{longtable} % tabelas longas
```

```
\usepackage[most]{tcolorbox} % caixas de texto
\usepackage{amsmath,amssymb}
\usepackage{hyperref}
% listings, fancyvrb e algorithm2e já vêm carregados pela classe coppe (v3.8).

\addbibresource{example.bib}% base de referências (biblatex/biber)

\makelosymbols
\makeloabbreviations

\begin{document}
  \title{Título da Tese}
  \foreigntitle{Thesis Title}
  \author{Nome do Autor}{Sobrenome}
  \advisor{Prof.}{Nome do Primeiro Orientador}{Sobrenome}{D.Sc.}
  \advisor{Prof.}{Nome do Segundo Orientador}{Sobrenome}{Ph.D.}
  \advisor{Prof.}{Nome do Terceiro Orientador}{Sobrenome}{D.Sc.}

  \examiner{Prof.}{Nome do Primeiro Examinador Sobrenome}{D.Sc.}
  \examiner{Prof.}{Nome do Segundo Examinador Sobrenome}{Ph.D.}
  \examiner{Prof.}{Nome do Terceiro Examinador Sobrenome}{D.Sc.}
  \examiner{Prof.}{Nome do Quarto Examinador Sobrenome}{Ph.D.}
  \examiner{Prof.}{Nome do Quinto Examinador Sobrenome}{Ph.D.}
  \department{PESC}
  \date{01}{2024}

  \keyword{Primeira palavra-chave}
  \keyword{Segunda palavra-chave}
  \keyword{Terceira palavra-chave}

  \maketitle

  \frontmatter
  \dedication{A alguém cujo valor é digno desta dedicatória.}

  \chapter*{Agradecimentos}

  Gostaria de agradecer a todos os que contribu\'iram para este trabalho:
  orientadores, colegas, familiares e \`as ag\^encias de fomento. Lorem ipsum
  dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt
  ut labore et dolore magna aliqua.

  \begin{abstract}

  Apresenta-se, nesta tese, um exemplo de resumo. Lorem ipsum dolor sit amet,
  consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
  dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
  ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
  in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
  pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
  officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
  iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
  aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
  vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
```

```
aspernatur aut odit aut fugit.

\end{abstract}

\begin{foreignabstract}

In this work, we present an example abstract. Lorem ipsum dolor sit amet,
consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
aspernatur aut odit aut fugit.

\end{foreignabstract}

% Ordem pre-textual (NBR 14724/6027, R55): listas (ilustracoes, tabelas,
% abreviaturas e siglas, simbolos) ANTES do sumario; sumario por ultimo.
\listoffigures
\listoftables
\listofquadros
\listofprogramas
\listofalgorithms
\printloabbreviations
\printlosymbols
\tableofcontents

\mainmatter
\chapter{Introdução}

Este é um documento exemplo para o uso da classe CoppeTeX, destinado a ajudar os alunos da

A classe \verb|coppe| foi criada por Vicente Helano e George Ainsworth, porém, em 2024, é

A versão mais atual dessa classe é mantida no GitHub, no repositório \url{https://github.c

Esse documento segue a norma de formatação de teses e dissertações da COPPE. Ele também po

Esse documento é usado como exemplo de coisas que podem ser feitas. Ele está configurado p

É importante de notar que essa classe não foi construída sobre a classe \LaTeX \  para a A

Apesar desse modelo ser muito bom, ele tem um defeito: a limitação do sistema de referênci

Mais ainda, as regras da COPPE ainda não se adaptaram, no início de 2024, as novas regras

Este documento não substitui, mas complementa, o documento que descreve a classe.
```

```
\chapter{Configurações Iniciais}

  A primeira coisa a fazer é escolher o tipo de documento. Isso é feito como uma opção no c

  Como pode ser visto nesse documento, muita coisa pode ser configurada, o que gerará o tra

  Recomendo ler o documento ``The \verb*|coppe| document class'' para entender melhor todas

\section{Linguagem principal do texto}

 Essa classe considera que o texto principal está em português e algumas partes específicas
\begin{itemize}
\item A opção \verb|english| deve ser usada no comando \verb|\documentclass|.
\item A bibliografia segue automaticamente o idioma do Babel (português ou inglês); não é p
\end{itemize}

 A variação de linguagem, em inglês ou português apenas, já é suportada pela classe Coppe\T


\section{Por que usar o \LaTeX}

 Há uma grande discussão entre usuário de Word e \LaTeX, principalmente, quanto ao uso dess

 Nós escolhemos o \LaTeX por alguns motivos: grande facilidade de seguir um estilo sem se p

 As principais desvantagens são: idiossincrasias que podem gastar tempo, pouco controle sob

\section{Como e onde usar o \LaTeX}

 Existem muitos tutoriais de \LaTeX, mas basicamente, em 2024, ele é usado em dois ambiente
 \begin{enumerate}
 \item Na sua máquina, instalando uma versão completa como o Mik\TeX, típico do Windows, ou
 \item Usar um ambiente na rede, como o Overleaf.
 \end{enumerate}

 Em todo caso, recomendo fortemente que, ao mesmo tempo, mantenha versões no Git e faça o b


\chapter{Algumas Regras da COPPE}

 Todas abreviaturas e símbolos devem ser definida antes de utilizada. Isso é facilmente fei

 É imprescindível definir os símbolos, tal como o
 conjunto dos números reais $\mathbb{R}$ e o conjunto vazio $\emptyset$.
 \symbl{$\mathbb{R}$}{Conjunto dos números reais}
 \symbl{$\emptyset$}{Conjunto vazio}. Usamos esse exemplo aqui justamente para mostrar como

 Para as listas de abreviaturas e símbolos funcionarem no Overleaf é necessário rodar o \ve

 Como as listas de símbolos e de abreviaturas usamo o mesmo comando usado para criar índice

\section{Citações}

Citações curtas podem ser feitas \quote{o comando quote} ou direto com ``duas crases e dois
```

```
  \begin{longquote}
  Um exemplo de citação longa nas regras da ABNT (4cm de recuo e fonte menor)
  feita com o ambiente  \verb=longquote= The primary objective of this
  investigation was to determine the feasibility of detecting corrosion in
  aluminum Naval aircraft components with neutron radiographic interrogation
  and the use of standard corrosion penetrameters. Secondary objectives
  included the determination of the effect of object thickness on image quality,
  the defining of minimum levels of detectability and a preliminary investigation
  of a means whereby the degree of corrosion could be quantified with neutron
  radiographic data. \cite{article-example}
  \end{longquote}

  Citações devem apontar as referências. Para isso, está disponível o ótimo pacote \verb*|na

  Em todo caso, \textbf{deve se tomar enorme atenção com as citações, para evitar ocorrer em

\chapter{Floats}

Grande parte dos problemas de iniciantes, e veteranos, em \LaTeX é da localização dos \texti

A regra geral de posicionamento é que uma figura ou quadro só pode aparecer a partir da mesm

\textbf{Pela norma atual (manual2024), a legenda} \verb|\caption| \textbf{de toda ilustração

Quadros são ilustrações cujos dados são delimitados por linhas em \emph{todas} as bordas (ao


\section{Tabelas e Figuras Padrão}

Vamos ver uma tabela padrão, como a \autoref{tab:exemplo_numeros}.

\begin{table}[ht]
\centering % Centraliza a tabela
\caption{Exemplo de Tabela de Números}
\label{tab:exemplo_numeros}
\begin{tabular}{ccc} % Define a quantidade de colunas
\hline % Linha superior
\textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ % Cabeçalhos
\hline % Linha média
1 & 2 & 3 \\ % Primeira linha de dados
\hline
4 & 5 & 6 \\ % Segunda linha de dados
\hline
7 & 8 & 9 \\ % Terceira linha de dados
\hline
10 & 11 & 12 \\ % Quarta linha de dados
\hline % Linha inferior
\end{tabular}
\fonte{Elabora\c{c}\~ao pr\'opria.} % fonte obrigatoria, abaixo do float
\end{table}
```

```
Já a \autoref{fig:exemplo_figura} é uma figura padrão, com controle da largura.

\begin{figure}[ht]
\centering % Centraliza a figura
\caption{Exemplo de Figura com Legenda Acima} % Legenda em cima (norma atual)
\label{fig:exemplo_figura} % Etiqueta para referência cruzada
\includegraphics[width=0.5\textwidth]{coppe-logo.pdf} % Inclui a imagem com metade da largur
\source{COPPE/UFRJ.} % fonte obrigatoria, abaixo da figura
\end{figure}

Quadros têm a mesma regra (legenda acima, fonte abaixo), mas com linhas em todas as bordas.

\begin{quadro}[ht]
\centering
\caption{Exemplo de Quadro}
\label{qua:exemplo}
\begin{tabular}{|l|l|}
\hline
\textbf{Ilustração} & \textbf{Delimitação} \\
\hline
Tabela & Linhas apenas no topo e na base. \\
\hline
Quadro & Linhas em todas as bordas. \\
\hline
\end{tabular}
\fonte{Elabora\c{c}\~ao pr\'opria.}
\end{quadro}

\section{Tabelas mais elegantes}

Atualmente a tendência é usar tabelas mais leves, como \autoref{tab:exemplo_numerosbom}. Iss

\begin{table}[ht]
\centering % Centraliza a tabela
\caption{Exemplo de Tabela de Números mais elegantes}
\label{tab:exemplo_numerosbom}
\begin{tabular}{ccc} % Define a quantidade de colunas
\toprule % Linha superior
\textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ % Cabeçalhos
\midrule % Linha média
1 & 2 & 3 \\ % Primeira linha de dados
4 & 5 & 6 \\ % Segunda linha de dados
7 & 8 & 9 \\ % Terceira linha de dados
10 & 11 & 12 \\ % Quarta linha de dados
\bottomrule % Linha inferior
\end{tabular}
\end{table}

\section{Tabelas Longas ou Largas}
```

```
Se sua tabela é muito longa ou larga, existem várias opções.
\begin{itemize}
    \item alterar o tamanho da letra
    \item Usar o longtable
    \item rodar a tabela, fazendo ela em \textit{landscape}
    \item fazer a tabela dentro de um minibox
\end{itemize}


\subsection{Tabelas largas demais}

É comum em teses que as tabelas sejam largas demais. Há várias formas de resolver isso.

A \autoref{tab:tabela_largafns} é larga demais, e nela isso é resolvido diminuindo a fonte p

\begin{table}[ht]
\centering % Centraliza a tabela
\caption{Exemplo de Tabela Larga com Fonte Menor}
\label{tab:tabela_largafns}
\footnotesize % Aplica uma fonte menor para a tabela
\begin{tabular}{cccccccc} % Aumente o número de colunas conforme necessário
\toprule
\textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} & \textbf{Coluna 4} & \textbf{Colu
\midrule
Dado 1.1 & Dado 1.2 & Dado 1.3 & Dado 1.4 & Dado 1.5 & Dado 1.6 & Dado 1.7 & Dado 1.8 \\
Dado 2.1 & Dado 2.2 & Dado 2.3 & Dado 2.4 & Dado 2.5 & Dado 2.6 & Dado 2.7 & Dado 2.8 \\
Dado 3.1 & Dado 3.2 & Dado 3.3 & Dado 3.4 & Dado 3.5 & Dado 3.6 & Dado 3.7 & Dado 3.8 \\
\bottomrule
\end{tabular}
\end{table}

O comando \verb|\resizebox{width}{height}{content}| permite ajustar o tamanho de qualquer co

\begin{table}[ht]
\centering
\caption{Exemplo de Tabela Redimensionada}
\label{tab:examplerb}
\resizebox{\textwidth}{!}{%
\begin{tabular}{llll}
\toprule
Coluna 1 & Coluna 2 & Coluna 3 & Coluna 4 \\
\midrule
Dados 1 & Dados 2 & Dados 3 & Dados 4 \\
Dados 5 & Dados 6 & Dados 7 & Dados 8 \\
\bottomrule
\end{tabular}%
}
\end{table}


Para rodar uma tabela muito larga em 90 graus no LaTeX, você pode usar o pacote \verb*|rotat

Aqui está um exemplo de como usar o ambiente \verb*|sidewaystable| para girar uma tabela. Pr

```

```
\begin{sidewaystable}
\centering
\caption{Sua Legenda Aqui}
\label{tab:sua_tabela}
\begin{tabular}{lll}
\toprule
Coluna 1 & Coluna 2 & Coluna 3 \\
\midrule
Item 1 & Item 2 & Item 3 \\
Item 4 & Item 5 & Item 6 \\
\bottomrule
\end{tabular}
\end{sidewaystable}

Se a tabela for muito longa, o ambiente \verb|longtable| é o ideal. Ele fornece comandos par

% Exemplo de tabela longa que se estende por várias páginas
\begin{longtable}{|c|c|c|}
% primeiro cabeçalho (é o caption)
\caption{Exemplo de Tabela Longa}\label{tab:longa} \\
\hline \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ \hline
\endfirsthead
% cabeçalho normal
\multicolumn{3}{c}%
{{\tablename\ \thetable{} -- continuação da página anterior}} \\
\hline \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ \hline
\endhead
% pé normal
\hline \multicolumn{3}{|r|}{{Continua na próxima página}} \\ \hline
\endfoot
\hline
% último pé
\multicolumn{3}{|r|}{{Continua na próxima página}}\\
\hline \hline
\endlastfoot

% Conteúdo da tabela
1 & 2 & 3 \\
4 & 5 & 6 \\
1 & 2 & 3 \\
4 & 5 & 6 \\
1 & 2 & 3 \\
4 & 5 & 6 \\
1 & 2 & 3 \\
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4 & 5 & 6 \\
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4 & 5 & 6 \\

% Repetir linhas semelhantes conforme necessário para estender a tabela por 3 páginas
\end{longtable}

  \chapter{Revis\~ao Bibliogr\'afica}

  Para ilustrar a completa ades\~ao ao estilo de cita{\c c}\~oes e listagem de
  refer\^encias bibliogr\'aficas, a Tabela~\ref{tab:citation} apresenta cita{\c
  c}\~oes de alguns dos trabalhos contidos na norma fornecida pela CPGP da
  COPPE, utilizando o estilo numérico. Tirando do comando inicial o parâmetro opcional numér

  \begin{table}[h]
  \caption{Exemplos de cita{\c c}\~oes utilizando o comando padr\~ao
    \texttt{\textbackslash cite} do \LaTeX\ e
    o comando \texttt{\textbackslash citet},
    fornecido pelo pacote \texttt{natbib}.}
  \label{tab:citation}
  \centering
```

```
  {\footnotesize
  \begin{tabular}{|c|c|c|}
    \hline
    Tipo da Publicação & \verb|\cite| & \verb|\citet|\\
    \hline
    Livro & \cite{book-example} & \citet{book-example}\\
    Artigo & \cite{article-example} & \citet{article-example}\\
    Relatório & \cite{techreport-example} & \citet{techreport-example}\\
    Relatório & \cite{techreport-exampleIn} & \citet{techreport-exampleIn}\\
    Anais de Congresso & \cite{inproceedings-example} &
      \citet{inproceedings-example}\\
    Séries & \cite{incollection-example} & \citet{incollection-example}\\
    Em Livro & \cite{inbook-example} & \citet{inbook-example}\\
    Dissertação de mestrado & \cite{mastersthesis-example} &
      \citet{mastersthesis-example}\\
    Tese de doutorado & \cite{phdthesis-example} & \citet{phdthesis-example}\\
    \hline
  \end{tabular}}
  \end{table}

 \begin{table}[h]
  \caption{Exemplos de cita{\c c}\~oes utilizando o comando padr\~ao
    \texttt{\textbackslash cite} do \LaTeX\ e
    o comando \texttt{\textbackslash citet},
    fornecido pelo pacote \texttt{natbib}. Além disso, usando o booktabs.}
  \label{tab:citation1}
  \centering
  {\footnotesize
  \begin{tabular}{ccc}
    \toprule
    Tipo da Publicação & \verb|\cite| & \verb|\citet|\\
    \midrule
    Livro & \cite{book-example} & \citet{book-example}\\
    Artigo & \cite{article-example} & \citet{article-example}\\
    Relatório & \cite{techreport-example} & \citet{techreport-example}\\
    Relatório & \cite{techreport-exampleIn} & \citet{techreport-exampleIn}\\
    Anais de Congresso & \cite{inproceedings-example} &
      \citet{inproceedings-example}\\
    Séries & \cite{incollection-example} & \citet{incollection-example}\\
    Em Livro & \cite{inbook-example} & \citet{inbook-example}\\
    Dissertação de mestrado & \cite{mastersthesis-example} &
      \citet{mastersthesis-example}\\
    Tese de doutorado & \cite{phdthesis-example} & \citet{phdthesis-example}\\
    \bottomrule
  \end{tabular}}
  \end{table}

\chapter{Alguns outros exemplo úteis}

\begin{tcolorbox}[title=Meu Textbox]
Este é o conteúdo do meu textbox. Você pode adicionar qualquer texto aqui, bem como incluir
\end{tcolorbox}

\begin{tcolorbox}
```

```
Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem co
\end{tcolorbox}

\begin{figure}[ht]
    \centering
    \begin{tikzpicture}
        \node[anchor=south west,inner sep=0] (image) at (0,0) {\includegraphics[width=0.5\te
        \begin{scope}[x={(image.south east)},y={(image.north west)}]
            % Definindo o textbox dentro da figura
            \node[anchor=north west, text width=0.3\textwidth, fill=white, opacity=0.7, text
                \begin{tcolorbox}[colback=red!5!white,colframe=red!75!black,title=Textbox de
                    Este textbox fala sobre como inserir um textbox dentro de uma figura usa
                \end{tcolorbox}
            };
        \end{scope}
    \end{tikzpicture}
    \caption{Figura com Textbox}
    \label{fig:figura_com_textbox1}
\end{figure}


\begin{figure}[ht]
    \centering
 \begin{tcolorbox}
Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem co
\end{tcolorbox}
    \caption{Figura com Textbox simples}
    \label{fig:figura_com_textbox}
\end{figure}

\section{Listagens de código e algoritmos}

O CoppeTeX traz embutido o suporte a listagens de programas (pacote \verb|listings|) e a alg

\begin{python}[caption={Exemplo de programa em Python},label=prog:exemplo]
def saudacao(nome):
    # imprime uma saudação acentuada
    print(f"Olá, {nome}!")

saudacao("COPPE")
\end{python}
\source{Elaboração própria.}

Há estilos prontos também para \verb|java|, \verb|xml|, \verb|html| e \verb|prolog|, comando

\begin{algorithm}[H]
\caption{Soma dos elementos de um vetor}
\label{alg:soma}
\Dados{vetor $v$ com $n$ elementos}
\Resultado{soma $s$ dos elementos de $v$}
$s \leftarrow 0$\;
\Para{$i \leftarrow 1$ \Ate{} $n$}{
  $s \leftarrow s + v[i]$\;
}
```

```
\Retorna $s$\;
\end{algorithm}

  \chapter{Método Proposto}
  \chapter{Resultados e Discuss\~oes}

  \section{Algumas Demonstra{\c c}\~oes}

  A  Lista de Símbolos precisa usar comandos específicos. Aqui vamos usar os símbolos $\alph
  \symbl[beta]{Beta}{A palavra Beta more e corrigida}
  \symbl[zzbeta]{$\beta$}{A letra $\beta$ corrigida}
  \symbl{beta}{A palavra beta}
  \symbl{alpha}{A palavra alpha}
  \symbl[alpha]{Alpha}{A palavra Alpha}
  \symbl[zzalpha]{$\alpha$}{A letra $\alpha$ corrigida}
  \symbl[marco]{Marco}{A palavra Marco corrigida}

  A Lista de Abreviações segue, a partir de 2024, a mesma regra, e aqui seguem alguns exempl
  \abbrev{GoT}{Game of Thrones}
  \abbrev[GOT]{GoT}{Game of Thrones ordenado como GOT}
  \abbrev[iot]{IoT}{IoT ordenado como iot}
  \abbrev[IoT]{IoT}{IoT ordenado como IoT}
  \abbrev[IOT]{IoT}{IoT ordenado como IOT}
  \abbrev{IoT}{IoT com ordenação default}
  \abbrev[ITU]{ITU}{ITU mesmo}

  \newsigla{ufrj}{UFRJ}{Universidade Federal do Rio de Janeiro}
  \newsigla{cpgp}{CPGP}{Comissão de Pós-Graduação e Pesquisa de Engenharia}

  As siglas (manual2024, seção 2.8) são expandidas na primeira ocorrência e registradas auto




  \chapter{Conclusões}

  \backmatter

  % Tipos exóticos da ABNT (NBR 6023): patente, legislação, norma, mapa,
  % partitura --- declarados via sinônimos em português em example.bib.
  \nocite{patente-example,legislacao-example,norma-example,mapa-example,partitura-example}
  \nocite{videogame-example,filme-example}
  \printbibliography



\appendix

\chapter{Um apêndice}

Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização

Apêndice: O apêndice é um texto ou documento elaborado pelo autor do trabalho com o objetivo

```

```

\chapter{Outro apêndice}

\annex


\chapter{Um Anexo}
Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilizaçã



Anexo: O anexo, por sua vez, consiste em um texto ou documento não elaborado pelo autor, que

No modelo \CoppeTeX os anexos devem obrigtoriamente vir depois dos apêndices e usam o comand


\chapter{Outro Anexo}


\end{document}

⟨/example⟩
```

# 8 Implementation

## 8.1 The 'coppe.cls' file

```
⟨*class⟩
\def\filename{coppe.dtx}
\def\fileversion{v3.8}
\def\filedate{2026/05/25}
\NeedsTeXFormat{LaTeX2e}[1995/12/01]
\ProvidesClass{coppe}[\filedate\ \fileversion\ COPPE Dissertations and Thesis]
\LoadClass[12pt,a4paper,twoside]{book}
% The bibliography engine (biblatex+biber) is loaded further below, after babel,
% so the numbers class option can select the citation style at load time.
\RequirePackage{hyphenat}
\RequirePackage{lastpage}
\RequirePackage{ifthen}
\RequirePackage{graphicx}
\RequirePackage{setspace}
\RequirePackage{tabularx}
\RequirePackage{etoolbox}
\RequirePackage{eqparbox}
\RequirePackage{ltxcmds}
\RequirePackage[T1]{fontenc}
% manual2024 margins: 3cm top, 2cm bottom, mirrored 3cm inner / 2cm outer for
% two-sided printing -- realised as inner=outer=2cm plus a 1cm bindingoffset.
\RequirePackage[a4paper,twoside,bindingoffset=1cm,%
top=3cm,bottom=2cm,inner=2cm,outer=2cm]{geometry}
\RequirePackage{titlesec}
% --- Section heading formats (manual2024 2.6 / proposta P22) -----------------
% chapter (numbered): "N  TITLE", ALL CAPS, bold, flush left (no "Capitulo N").
% APENDICE/ANEXO chapters print "APENDICE A -- Title" (R75/R76); normal chapters
```

```
% keep "N Title". \appendix and \annex raise the flag (see below).
\newif\if@coppeappendix \@coppeappendixfalse
\titleformat{\chapter}[hang]
  {\normalfont\Large\bfseries}
  {\if@coppeappendix\MakeUppercase{\appendixname}\ \thechapter\ \textendash\else\thechapter\
  {1ex}{\MakeUppercase}
\titlespacing*{\chapter}{0pt}{10pt}{1.5\baselineskip}
\let\coppe@orig@appendix\appendix
\renewcommand\appendix{\coppe@orig@appendix\@coppeappendixtrue}
% chapter (unnumbered): centred ALL CAPS bold (errata, lists, resumo, sumario,
% referencias, glossario, ... -- unnumbered headings are centred, P29-P30).
\titleformat{name=\chapter,numberless}[hang]
  {\normalfont\Large\bfseries\filcenter}{}{0pt}{\MakeUppercase}
\titlespacing*{name=\chapter,numberless}{0pt}{10pt}{1.5\baselineskip}
% section: ALL CAPS, normal weight.
\titleformat{\section}
  {\normalfont\large\bfseries}{\thesection}{1ex}{\MakeUppercase}
% subsection: bold (author writes Title Case).
\titleformat{\subsection}
  {\normalfont\bfseries}{\thesubsection}{1ex}{}
% subsubsection: bold italic.
\titleformat{\subsubsection}
  {\normalfont\bfseries\itshape}{\thesubsubsection}{1ex}{}
% --- Captions (manual2024 2.10-2.11 / proposta P6): smaller font, ABNT
%     "Figura 1 -- titulo" travessao separator (sources handled by \source). ---
\RequirePackage{caption}
\captionsetup{font=footnotesize,labelsep=endash}
% --- Floats: caption on TOP and a mandatory "Fonte:" line at the bottom.
%     manual2024 2.10-2.11 (R46/R51: identification on top; R48/R52: the source
%     at the bottom is mandatory). The float package's plaintop style forces the
%     caption above the body for figure, table and the new quadro float. -------
\RequirePackage{float}
\floatstyle{plaintop}
\restylefloat{figure}
\restylefloat{table}
% New "quadro" float -- "Quadro" (pt) / "Frame" (en) (R53) -- with its own list
% file (.loq). A quadro's data is bordered on all sides (drawn by the author),
% unlike a tabela (rules on top and bottom only).
\providecommand{\quadroname}{\iflanguage{brazilian}{Quadro}{Frame}}
\providecommand{\listquadroname}{%
  \iflanguage{brazilian}{Lista de Quadros}{List of Frames}}
\newfloat{quadro}{htbp}{loq}[chapter]
\floatname{quadro}{\quadroname}
\providecommand{\quadroautorefname}{\quadroname}
% Format the Lista de Quadros entries exactly like the Lista de Figuras ones
% (float's own \l@quadro lays the number/title out incorrectly).
\let\l@quadro\l@figure
\newcommand\listofquadros{%
  \chapter*{\listquadroname}%
  \addcontentsline{toc}{chapter}{\listquadroname}%
  \@mkboth{\MakeUppercase\listquadroname}{\MakeUppercase\listquadroname}%
  \@starttoc{loq}}
% \source / \fonte: the illustration source (ABNT-mandatory). Typeset below the
% float as a smaller, single-spaced, centred line; it does NOT advance any
```

```
% caption counter. \cpsource/\cpfonte are clash-safe aliases (the rename rule);
% the label is "Fonte" (pt) / "Source" (en).
\newcommand{\cpsourcename}{\iflanguage{brazilian}{Fonte}{Source}}
\newcommand{\cpsource}[1]{%
  \par\addvspace{\smallskipamount}%
  {\footnotesize\singlespacing\centering\cpsourcename: #1\par}}
\let\cpfonte\cpsource
\@ifundefined{source}{\let\source\cpsource}{}%
\@ifundefined{fonte}{\let\fonte\cpsource}{}%
% \newcoppefloat{env}{caption name}{list title}: declare an extra caption-on-top
% float (e.g. "picture") that supports \source and gains a \listof<env>s command.
\newcommand{\newcoppefloat}[3]{%
  {\floatstyle{plaintop}\newfloat{#1}{htbp}{lo#1}}%
  \floatname{#1}{#2}%
  \expandafter\newcommand\csname listof#1s\endcsname{%
    \chapter*{#3}\addcontentsline{toc}{chapter}{#3}%
    \@mkboth{\MakeUppercase{#3}}{\MakeUppercase{#3}}%
    \@starttoc{lo#1}}}
% --- Footnotes (manual2024 / R16, R8, R21): single spacing and a 5 cm rule
%     (filete) from the left margin; footnotesize is the book default. --------
\let\coppe@orig@footnotetext\@footnotetext
\long\def\@footnotetext#1{\coppe@orig@footnotetext{\singlespacing#1}}
\renewcommand\footnoterule{\kern-3\p@\hrule\@width 5cm\kern 2.6\p@}
% --- alineas: list keyed by lowercase letters a) b) c); subitems with "--"
%     (manual2024 2.6 / proposta P24). ---------------------------------------
\RequirePackage{enumitem}
\newlist{alineas}{enumerate}{2}
\setlist[alineas,1]{label=\alph*),leftmargin=1.8em,itemsep=0pt}
\setlist[alineas,2]{label=--,leftmargin=1.8em,itemsep=0pt}
% --- Page numbering: arabic in the top OUTER corner on textual pages
%     (manual2024 2.7 / proposta P25-P26). Pre-textual numbering is set by
%     \frontmatter (unnumbered by default; roman with the numeraisromanos
%     class option). ---------------------------------------------------------
\RequirePackage{fancyhdr}
\fancypagestyle{coppe}{\fancyhf{}\fancyhead[RO,LE]{\thepage}%
  \renewcommand{\headrulewidth}{0pt}}
% Blank pages inserted by \cleardoublepage (two-sided) must be empty: no page
% number may leak onto them (e.g. the verso right after the cover).
\renewcommand{\cleardoublepage}{%
  \clearpage
  \if@twoside
    \ifodd\c@page\else
      \hbox{}\thispagestyle{empty}\newpage
      \if@twocolumn\hbox{}\newpage\fi
    \fi
  \fi}
\def\CoppeTeX{{\rm C\kern-.05em{\sc o\kern-.025em p\kern-.025em
p\kern-.025em e}}\kern-.08em
T\kern-.1667em\lower.5ex\hbox{E}\kern-.125emX\spacefactor1000}

\newboolean{maledoc}
\setboolean{maledoc}{false}
%
% Class options.
% If you are writing a text in English, you must turn ``English'' on.
```

```
% Otherwise, Portuguese is considered the main language.
\newif\if@english\@englishfalse
\DeclareOption{english}{\@englishtrue}
\DeclareOption{msc}{%
  \newcommand{\@degree}{M.Sc.}
  \newcommand{\@degreename}{Mestrado}
  \newcommand{\local@degname}{Mestre}
  \newcommand{\foreign@degname}{Master}
  \newcommand\local@doctype{Disserta{\c c}{\~ a}o}
  \newcommand\foreign@doctype{Dissertation}
}
\DeclareOption{dscexam}{%
  \newcommand{\@degree}{D.Sc.}
  \newcommand{\@degreename}{Doutorado}
  \newcommand{\local@degname}{Doutor}
  \newcommand{\foreign@degname}{Doctor}
  \setboolean{maledoc}{true}
  \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
  \newcommand\foreign@doctype{Qualifying Exam}
}
\DeclareOption{mscexam}{%
 \newcommand{\@degree}{M.Sc.}
  \newcommand{\@degreename}{Mestrado}
  \newcommand{\local@degname}{Mestre}
  \newcommand{\foreign@degname}{Master}
  \setboolean{maledoc}{true}
  \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
  \newcommand\foreign@doctype{Qualifying Exam}
}
\DeclareOption{dsc}{%
  \newcommand{\@degree}{D.Sc.}
  \newcommand{\@degreename}{Doutorado}
  \newcommand{\local@degname}{Doutor}
  \newcommand{\foreign@degname}{Doctor}
  \newcommand\local@doctype{Tese}
  \newcommand\foreign@doctype{Thesis}
}
\newif\if@coppenumeric\@coppenumericfalse
\DeclareOption{numbers}{\@coppenumerictrue}
% numeraisromanos: show roman numerals on the pre-textual pages. Default: the
% pre-textual pages are counted but not numbered (manual2024 2.7).
\newif\if@copperoman\@copperomanfalse
\DeclareOption{numeraisromanos}{\@copperomantrue}
```

Here is the default one-and-a-half line spacing. Users can change to double spacing by passing the `doublespacing` option.

```
\onehalfspacing
\DeclareOption{doublespacing}{%
  \doublespacing
}
\ProcessOptions\relax
\if@english
  \RequirePackage[brazilian,english]{babel}
\else
```

```
  \RequirePackage[english,brazilian]{babel}
\fi
% Quotation marks: csquotes gives the language-aware \enquote; \citacao is its
% Portuguese alias for short, inline quotations (proposta P3).
\RequirePackage{csquotes}
\newcommand{\citacao}[1]{\enquote{#1}}
% --- Code listings & algorithms (TODO3 "tratar listagens"): the listings
%     package gives the "Programa" listing (caption on top, ABNT R46) with
%     \listofprogramas; algorithm2e gives algorithms. Names are babel-aware and
%     the language styles below are based on the user’s listagens.tex. ---------
\RequirePackage{xcolor}
\RequirePackage{listings}
\RequirePackage{fancyvrb}
\definecolor{deepblue}{rgb}{0,0,0.5}
\definecolor{deepred}{rgb}{0.6,0,0}
\definecolor{deepgreen}{rgb}{0,0.5,0}
\DeclareFixedFont{\ttb}{T1}{txtt}{bx}{n}{10}
\DeclareFixedFont{\ttm}{T1}{txtt}{m}{n}{10}
\newcommand{\gxnumbersep}{0.5em}
\lstset{captionpos=t,extendedchars=true,basicstyle=\ttfamily,inputencoding=utf8,%
  literate=%
  {ã}{{\~{a}}}1 {á}{{\’{a}}}1 {â}{{\^{a}}}1 {à}{{\‘{a}}}1 {é}{{\’{e}}}1%
  {è}{{\‘{e}}}1 {ê}{{\^{e}}}1 {î}{{\^{i}}}1 {í}{{\’{i}}}1 {ô}{{\^{o}}}1%
  {ó}{{\’{o}}}1 {õ}{{\~{o}}}1 {û}{{\^{u}}}1 {ú}{{\’{u}}}1 {ç}{{\c{c}}}1%
  {Ç}{{\c{C}}}1 {À}{{\‘{A}}}1 {Â}{{\^{A}}}1 {Á}{{\’{A}}}1 {É}{{\’{E}}}1%
  {Ê}{{\^{E}}}1 {Í}{{\’{I}}}1 {Ó}{{\’{O}}}1 {Ô}{{\^{O}}}1 {Õ}{{\~{O}}}1%
  {Ú}{{\’{U}}}1%
}
% Babel-aware listing names, set after listings’ own babel setup.
\AtBeginDocument{%
  \renewcommand\lstlistingname{\iflanguage{brazilian}{Programa}{Program}}%
  \renewcommand\lstlistlistingname{%
    \iflanguage{brazilian}{Lista de Programas}{List of Programs}}}
% \listofprogramas in the coppe list style (heading + ToC + running head).
\newcommand\listofprogramas{%
  \chapter*{\lstlistlistingname}%
  \addcontentsline{toc}{chapter}{\lstlistlistingname}%
  \@mkboth{\MakeUppercase\lstlistlistingname}{\MakeUppercase\lstlistlistingname}%
  \@starttoc{lol}}
% Per-language styles + environments (python, java, xml, html, prolog).
\newcommand\pythonstyle{\lstset{language=Python,basicstyle=\ttm,numbers=left,%
  numberstyle=\small\ttfamily,numbersep=\gxnumbersep,commentstyle=\color{purple},%
  morekeywords={self,as},keywordstyle=\ttb\color{deepblue},%
  emph={MyClass,__init__},emphstyle=\ttb\color{deepred},%
  stringstyle=\color{deepgreen},frame=tb,showstringspaces=false,%
  extendedchars=true,firstnumber=auto,inputencoding=utf8,breaklines=true,%
  escapechar=‘,postbreak=\mbox{\textcolor{red}{$\hookrightarrow$}\space}}}
\lstnewenvironment{python}[1][]{\pythonstyle\lstset{#1}%
  \csname\@lst @SetFirstNumber\endcsname}{\csname\@lst @SaveFirstNumber\endcsname}
\newcommand\pythoninline[1]{{\pythonstyle\lstinline!#1!}}
\newcommand\pythonexternal[2][]{{\pythonstyle\lstinputlisting[#1]{#2}}}
\newcommand\xmlstyle{\lstset{language=XML,basicstyle=\ttm,numbers=left,%
  numberstyle=\small\ttfamily,numbersep=\gxnumbersep,%
  morekeywords={xs:schema,xs:element,xs:sequence,xs:complexType},%
```

```
  keywordstyle=\ttb\color{deepblue},%
  emph={name,type,attributeFormDefault,elementFormDefault,xmlns:xs},%
  emphstyle=\ttb\color{deepred},stringstyle=\color{deepgreen},frame=tb,%
  showstringspaces=false,extendedchars=true,inputencoding=utf8,breaklines=true,%
  postbreak=\mbox{\textcolor{red}{$\hookrightarrow$}\space}}}
\lstnewenvironment{xml}[1][]{\xmlstyle\lstset{#1}}{}
\newcommand\htmlstyle{\lstset{language=HTML,basicstyle=\ttm,numbers=left,%
  numberstyle=\small\ttfamily,numbersep=\gxnumbersep,%
  keywordstyle=\ttb\color{deepblue},emphstyle=\ttb\color{deepred},%
  stringstyle=\color{deepgreen},frame=tb,showstringspaces=false,%
  extendedchars=true,inputencoding=utf8,breaklines=true,%
  postbreak=\mbox{\textcolor{red}{$\hookrightarrow$}\space}}}
\lstnewenvironment{html}[1][]{\htmlstyle\lstset{#1}}{}
\newcommand\prologstyle{\lstset{language=Prolog,basicstyle=\ttm,numbers=left,%
  numberstyle=\small\ttfamily,numbersep=\gxnumbersep,commentstyle=\color{purple},%
  keywordstyle=\ttb\color{deepblue},emph={MyClass,__init__},%
  emphstyle=\ttb\color{deepred},stringstyle=\color{deepgreen},frame=tb,%
  showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
  breaklines=true,escapechar=`,%
  postbreak=\mbox{\textcolor{red}{$\hookrightarrow$}\space}}}
\lstnewenvironment{prolog}[1][]{\prologstyle\lstset{#1}%
  \csname\@lst @SetFirstNumber\endcsname}{\csname\@lst @SaveFirstNumber\endcsname}
\newcommand\javastyle{\lstset{language=Java,basicstyle=\ttm,numbers=left,%
  numberstyle=\small\ttfamily,numbersep=\gxnumbersep,commentstyle=\color{purple},%
  morekeywords={self,as},keywordstyle=\ttb\color{deepblue},%
  emphstyle=\ttb\color{deepred},stringstyle=\color{deepgreen},frame=tb,%
  showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
  breaklines=true,escapechar=`,%
  postbreak=\mbox{\textcolor{red}{$\hookrightarrow$}\space}}}
\lstnewenvironment{java}[1][]{\javastyle\lstset{#1}%
  \csname\@lst @SetFirstNumber\endcsname}{\csname\@lst @SaveFirstNumber\endcsname}
\newcommand\javainline[1]{{\javastyle\lstinline!#1!}}
\newcommand\javaexternal[2][]{{\javastyle\lstinputlisting[#1]{#2}}}
% Verbatim output environments (program output), fancyvrb.
\DefineVerbatimEnvironment{gxoutput}{Verbatim}%
  {frame=lines,numbers=left,numbersep=\gxnumbersep}
\DefineVerbatimEnvironment{gxoutputs}{Verbatim}%
  {frame=lines,numbers=left,numbersep=\gxnumbersep,fontsize=\small}
\renewcommand{\theFancyVerbLine}{{\small\ttfamily\arabic{FancyVerbLine}}}
% Algorithms: algorithm2e, language following the english class option.
\if@english
  \RequirePackage[linesnumbered,lined,algochapter,ruled]{algorithm2e}
\else
  \RequirePackage[portuguese,linesnumbered,lined,algochapter,ruled]{algorithm2e}
\fi
\SetKwFor{Para}{para}{faça}{fim do para}
\SetKwFor{Enquanto}{enquanto}{faça}{fim do enquanto}
% Bibliography engine: the full coppe BibLaTeX/biber ABNT style, replacing the
% former natbib+bibtex setup. Loaded here, after babel, so its language strings
% map correctly. The numbers option selects the numeric style; otherwise the
% author-date default is used. natbib=true keeps \citet/\citep available.
\if@coppenumeric
  \RequirePackage[backend=biber,datamodel=coppe,%
    bibstyle=coppe-numeric,citestyle=coppe-numeric,natbib=true]{biblatex}
```

```
\else
  \RequirePackage[backend=biber,datamodel=coppe,%
    bibstyle=coppe,citestyle=coppe,natbib=true]{biblatex}
\fi
% Reference-list heading "Referencias" (pt) / "References" (en) -- manual2024
% 3.1.4.1 requires the post-textual list to be titled "Referencias", not
% "Bibliografia" (babel's default \bibname). The title is set directly with
% \iflanguage so it is correct in BOTH the chapter heading and the .toc entry:
% \iflanguage resolves to plain text and survives the write to the .toc, whereas
% biblatex's \bibstring does not, and babel's \bibname caption override proved
% unreliable here. Evaluated in the body, the current language is settled.
\defbibheading{bibliography}{%
  \chapter*{\iflanguage{brazilian}{Refer\^encias}{References}}%
  \addcontentsline{toc}{chapter}{\iflanguage{brazilian}{Refer\^encias}{References}}}
% The catalog page range (bib:begin/bib:end) was written by the old
% thebibliography redefinition, which biblatex's \printbibliography does not
% use. biblatex offers \AtBeginBibliography but no end counterpart, so the
% begin label uses that hook and the end label is emitted by wrapping
% \printbibliography to write it just after the reference list.
\AtBeginBibliography{\coppe@bibBegin}
\let\coppe@orig@printbibliography\printbibliography
\renewcommand\printbibliography[1][]{%
  \coppe@orig@printbibliography[#1]%
  \coppe@bibEnd}
```

`\department` This macro is used to set the author's affiliation. There are twelve options which correspond to all academic units at COPPE/UFRJ. It defines the current and the foreign names of these units.

```
\newcommand\department[1]{%
  \ifthenelse{\equal{#1}{PEB}}
    {\global\def\local@deptname{Engenharia Biom{\' e}dica}
     \global\def\foreign@deptname{Biomedical Engineering}}{}
  \ifthenelse{\equal{#1}{PEC}}
    {\global\def\local@deptname{Engenharia Civil}
     \global\def\foreign@deptname{Civil Engineering}}{}
  \ifthenelse{\equal{#1}{PEE}}
    {\global\def\local@deptname{Engenharia El{\' e}trica}
     \global\def\foreign@deptname{Electrical Engineering}}{}
  \ifthenelse{\equal{#1}{PEM}}
    {\global\def\local@deptname{Engenharia Mec{\^ a}nica}
     \global\def\foreign@deptname{Mechanical Engineering}}{}
  \ifthenelse{\equal{#1}{PEMM}}
    {\global\def\local@deptname{Engenharia Metal{\' u}rgica e de Materiais}
\global\def\foreign@deptname{Metallurgical and Materials Engineering}}{}
  \ifthenelse{\equal{#1}{PEN}}
    {\global\def\local@deptname{Engenharia Nuclear}
     \global\def\foreign@deptname{Nuclear Engineering}}{}
  \ifthenelse{\equal{#1}{PENO}}
    {\global\def\local@deptname{Engenharia Oce{\^ a}nica}
     \global\def\foreign@deptname{Ocean Engineering}}{}
  \ifthenelse{\equal{#1}{PPE}}
    {\global\def\local@deptname{Planejamento Energ{\' e}tico}
     \global\def\foreign@deptname{Energy Planning}}{}
  \ifthenelse{\equal{#1}{PEP}}
```

```
    {\global\def\local@deptname{Engenharia de Produ{\c c}{\~ a}o}
     \global\def\foreign@deptname{Production Engineering}}{}
  \ifthenelse{\equal{#1}{PEQ}}
    {\global\def\local@deptname{Engenharia Qu{\' i}mica}
     \global\def\foreign@deptname{Chemical Engineering}}{}
  \ifthenelse{\equal{#1}{PESC}}
    {\global\def\local@deptname{Engenharia de Sistemas e Computa{\c c}{\~ a}o}
     \global\def\foreign@deptname{Systems Engineering and Computer Science}}{}
  \ifthenelse{\equal{#1}{PET}}
    {\global\def\local@deptname{Engenharia de Transportes}
     \global\def\foreign@deptname{Transportation Engineering}}{}
  \ifthenelse{\equal{#1}{PENT}}
    {\global\def\local@deptname{Engenharia de Nanotecnologia}
      \global\def\foreign@deptname{Nanotechnology Engineering}}{}
}
```

`\title` Used to enter the title in Brazilian Portuguese.

```
\renewcommand\title[1]{%
  \global\def\local@title{#1}%
}
```

`\foreigntitle` Used to enter the foreign title.

```
\newcommand\foreigntitle[1]{%
  \global\def\foreign@title{#1}%
}
```

`\advisor` Defines globally the title, name and academic degree of the advisors.

```
\newcount\@advisor\@advisor0
\newcommand\advisor[4]{%
  \global\@namedef{CoppeAdvisorTitle:\expandafter\the\@advisor}{#1}
  \global\@namedef{CoppeAdvisorName:\expandafter\the\@advisor}{#2}
  \global\@namedef{CoppeAdvisorSurname:\expandafter\the\@advisor}{#3}
  \global\@namedef{CoppeAdvisorDegree:\expandafter\the\@advisor}{#4}
  \global\advance\@advisor by 1
  \ifnum\@advisor>1
    \renewcommand\local@advisorstring{Orientadores}
    \renewcommand\foreign@advisorstring{Advisors}
  \fi
}
```

`\examiner`

```
\newcount\@examiner\@examiner0
\newcommand\examiner[3]{%
  \global\@namedef{CoppeExaminer:\expandafter\the\@examiner}{#1\ #2}
  \global\advance\@examiner by 1
}
```

`\author` It was redefined to allow the identification of the author's first names and surname.

```
\renewcommand\author[2]{%
  \global\def\@authname{#1}
  \global\def\@authsurn{#2}
}
```

\date This code makes easy to switch from dates in different languages.

```
\renewcommand\date[2]{%
  \month=#1
  \year=#2
}
```

\local@monthname

```
\newcommand\local@monthname{\ifcase\month\or
  Janeiro\or Fevereiro\or Mar{\c c}o\or Abril\or Maio\or Junho\or
  Julho\or Agosto\or Setembro\or Outubro\or Novembro\or Dezembro\fi}
```

\foreign@monthname

```
\newcommand\foreign@monthname{\ifcase\month\or
  January\or February\or March\or April\or May\or June\or
  July\or August\or September\or October\or November\or December\fi}
```

\keyword

```
\newcounter{keywords}
\newcommand\keyword[1]{%
  \global\@namedef{CoppeKeyword:\expandafter\the\c@keywords}{#1}
  \global\addtocounter{keywords}{1}
}
```

\freeconfig This command allows easy changing of core class parameters.

```
\newcommand\freeconfig[6]{%
 \providecommand\@degree{}
 \renewcommand\@degree{#1}
 \providecommand\@degreename{}
 \renewcommand\@degreename{#2}
 \providecommand\local@degname{}
 \renewcommand\local@degname{#3}
 \providecommand\foreign@degname{}
 \renewcommand\foreign@degname{#4}
 \providecommand\local@doctype{}
 \renewcommand\local@doctype{#5}
 \providecommand\foreign@doctype{}
 \renewcommand\foreign@doctype{#6}%
}
%    \end{macrocode}
% \end{macro}
%
% \begin{macro}{\frontmatter}
% The number of pages for both frontmatter and mainmatter printed
% in the cataloging details page is computed by means of simple
% \LaTeX\ labels.
%    \begin{macrocode}
\renewcommand\frontmatter{%
  \cleardoublepage
  \@mainmatterfalse
  \pagenumbering{roman}
  \thispagestyle{empty}
  % folha de rosto = counted page 1 (ABNT: the capa is not counted). An odd
  % start keeps the page-counter parity aligned with the physical (recto) side,
```

```
  % so \cleardoublepage lands every pre-textual element on an odd (front) page.
  \setcounter{page}{1}
  \makefrontpage
  \clearpage
  \if@copperoman
    \fancypagestyle{plain}{\fancyhf{}\fancyhead[RO,LE]{\thepage}%
      \renewcommand{\headrulewidth}{0pt}}%
    \pagestyle{coppe}%
  \else
    \fancypagestyle{plain}{\fancyhf{}\renewcommand{\headrulewidth}{0pt}}%
    \pagestyle{empty}%
  \fi
  \ifthenelse{\boolean{maledoc}}{}{\makecatalog}%
}
```

\mainmatter

```
\renewcommand\mainmatter{%
  \coppe@mainBegin
  \cleardoublepage
  \@mainmattertrue
  \fancypagestyle{plain}{\fancyhf{}\fancyhead[RO,LE]{\thepage}%
    \renewcommand{\headrulewidth}{0pt}}%
  \pagestyle{coppe}%
  \pagenumbering{arabic}}
```

\backmatter

```
\renewcommand\backmatter{%
  \if@openright
    \cleardoublepage
  \else
    \clearpage
  \fi}
%
```

\maketitle

```
\renewcommand\maketitle{%
  \pagenumbering{alph}
  \ltx@ifpackageloaded{hyperref}{\coppe@hypersetup}{}%
  \begin{titlepage}
  \begin{flushleft}
  \vspace*{1.5mm}
  \setlength\baselineskip{0pt}
  \setlength\parskip{1mm}
  \makebox[\linewidth]{%
    \includegraphics[height=2cm]{ufrj-logo}\hfill
    \includegraphics[height=2cm]{coppe-logo}}
  \end{flushleft}
  \vspace{1.05cm}
  \begin{center}
  \nohyphens{%
    \if@english
      \MakeUppercase\foreign@title
    \else
      \MakeUppercase\local@title
```

```
    \fi}\par
  \vspace*{3cm}
  \nohyphens{\@authname\ \@authsurn}\par
  \end{center}
  \vspace*{2.1cm}
  \begin{flushright}
  \begin{minipage}{8.45cm}
  \frontcover@maintext
  \end{minipage}\par
  \vspace*{7.5mm}
  \nohyphens{%
  \begin{tabularx}{8.45cm}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
    \local@advisorstring: &
    \count1=0
    \toks@={}
    \@whilenum \count1<\@advisor \do{%
    \ifcase\count1 % same as \ifnum0=\count1
      \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
        \expandafter\endcsname\expandafter\space%
        \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
    \else
      \toks@=\expandafter\expandafter\expandafter{%
        \expandafter\the\expandafter\toks@%
        \expandafter&\expandafter\space%
        \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
        \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
      }%
    \fi
    \advance\count1 by 1}
    \the\toks@
  \end{tabularx}}\par
  \end{flushright}
  \vspace*{\fill}
  \begin{center}
  \local@cityname\par
  \local@monthname\ de \number\year
  \end{center}
  \end{titlepage}
  \global\let\maketitle\relax%
  \global\let\and\relax}

\newcommand\makefrontpage{%
  \begin{center}
    \sloppy\nohyphens{
      \if@english
        \MakeUppercase\foreign@title
      \else
        \MakeUppercase\local@title
      \fi}\par
    \vspace*{7mm}
    {\@authname\ \@authsurn}\par
  \end{center}\par
  \vspace*{4mm}
  \frontpage@maintext
```

```
  \vspace*{16mm}
  \nohyphens{%
  \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
    \local@advisorstring: &
    \count1=0
    \toks@={}
    \@whilenum \count1<\@advisor \do{%
    \ifcase\count1 % same as \ifnum0=\count1
      \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
        \expandafter\endcsname\expandafter\space%
        \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
    \else
      \toks@=\expandafter\expandafter\expandafter{%
        \expandafter\the\expandafter\toks@%
        \expandafter&\expandafter\space%
        \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
        \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
      }%
    \fi
    \advance\count1 by 1}
    \the\toks@
  \end{tabularx}\par
  \vspace*{20mm}
  \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
    Aprovada por: &
    \count1=0
    \toks@={}
    \@whilenum \count1<\@examiner \do{%
    \ifcase\count1 % same as \ifnum0=\count1
      \toks@=\expandafter{\csname CoppeExaminer:\the\count1%
        \expandafter\endcsname\expandafter\\}
    \else
      \toks@=\expandafter\expandafter\expandafter{%
        \expandafter\the\expandafter\toks@%
        \expandafter&\expandafter\space%
        \csname CoppeExaminer:\the\count1\expandafter\endcsname%
        \expandafter\space\\
      }%
    \fi
    \advance\count1 by 1}
    \the\toks@
    \end{tabularx}}\par
  \vspace*{\fill}
  \frontpage@bottomtext}

\newcommand\coppe@hypersetup{%
\begingroup
  % changes to \toks@ and \count@ are kept local;
  % it's not necessary for them, but it is usually the case
  % for \count1, because the first ten counters are written
  % to the DVI file, thus you got lucky because of PDF output
  \toks@={}% in this special case not necessary
  \count@=0 %
  \@whilenum\count@<\value{keywords}\do{%
```

```
    % * a keyword separator is not necessary,
    %    if there is just one keyword
    % * \csname CoppeKeyword:\the\count@\endcsname must be expanded
    %    at least once, to get rid of the loop depended \count@
    \ifcase\count@ % same as \ifnum0=\count@
      \toks@=\expandafter{\csname CoppeKeyword:\the\count@\endcsname}%
    \else
      \toks@=\expandafter\expandafter\expandafter{%
        \expandafter\the\expandafter\toks@
        \expandafter;\expandafter\space
        \csname CoppeKeyword:\the\count@\endcsname
      }%
    \fi
    \advance\count@ by 1 %
  }%
  \edef\x{\endgroup
    \noexpand\hypersetup{%
      pdfkeywords={\the\toks@}%
    }%
  }%
\x
\hypersetup{%
  pdfauthor={\@authname\ \@authsurn},
  pdftitle={\local@title},
  pdfsubject={\local@doctype\ de \@degreename\ em \local@deptname\ da COPPE/UFRJ},
  pdfcreator={LaTeX with CoppeTeX toolkit},
  breaklinks={true},
  raiselinks={true},
  pageanchor={true},
}}
```

`\makecatalog` When the document has illustrations, it is required to insert ": il.;" between the number of pages of the textual part and the page dimension. We have created a label to flag the existence of lists of figures. It is checked to be undefined using the plain TEX command `\@isundefined` [?].

```
\newcommand\makecatalog{%
  \vspace*{\fill}
  \begin{center}
    \setlength{\fboxsep}{5mm}
    \framebox[120mm][c]{\makebox[5mm][c]{}%
      \begin{minipage}[c]{105mm}
      \setlength{\parindent}{5mm}
      \noindent\sloppy\nohyphens\@authsurn,
      \nohyphens\@authname\par
      \nohyphens{%
        \if@english
          \foreign@title%
        \else
          \local@title%
        \fi/\@authname\ \@authsurn. -- \local@cityname:
      UFRJ/COPPE, \number\year.}\par
      \pageref{front:pageno},
      \pageref{LastPage}
      p.\@ifundefined{r@cat:lofflag}{}{\pageref{cat:lofflag}} $29,7$cm.\par
```

```
    % There is an issue here. When the last entry must be split between lines,
    % the spacing between it and the next paragraph becomes smaller.
    % Should we manually introduce a fixed space? But how could we know that
    % a name was split? Is this happening yet?
    \nohyphens{%
    \begin{tabularx}{100mm}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
      \local@advisorstring: &
      \count1=0
      \toks@={}
      \@whilenum \count1<\@advisor \do{%
      \ifcase\count1 % same as \ifnum0=\count1
        \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
        \expandafter\endcsname\expandafter\space%
        \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
      \else
        \toks@=\expandafter\expandafter\expandafter{%
          \expandafter\the\expandafter\toks@
          \expandafter&\expandafter\space
          \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
          \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
        }%
      \fi
      \advance\count1 by 1}
      \the\toks@
    \end{tabularx}}\par
    \nohyphens{\local@doctype\ ({\MakeLowercase\@degreename}) --
    UFRJ/COPPE/Programa de \local@deptname, \number\year.}\par
    Refer{\^ e}ncias Bibliogr{\' a}ficas: p. \pageref{bib:begin} -- \pageref{bib:end}.\par
    \count1=0
    \count2=1
    \nohyphens{\@whilenum \count1<\value{keywords} \do {%
      \number\count2. \csname CoppeKeyword:\the\count1 \endcsname.
      \advance\count1 by 1
      \advance\count2 by 1}
    I. \csname CoppeAdvisorSurname:0\endcsname,%
    \ \csname CoppeAdvisorName:0\endcsname%
    \ifthenelse{\@advisor>1}{\ \emph{et~al.}{}}.
    II. \local@universityname, COPPE, Programa de \local@deptname.
    III. T{\' i}tulo.}
  \end{minipage}}
\end{center}
\vspace*{\fill}}
```

\dedication

```
\newcommand\dedication[1]{
\gdef\@dedic{#1}
  \cleardoublepage
  \vspace*{\fill}
  \begin{flushright}
    \begin{minipage}{60mm}
      \raggedleft \it \normalsize \@dedic
    \end{minipage}
  \end{flushright}}
```

`abstract` (*env.*) This is a specialization of the abstract in the article standard class.

```
\newenvironment{abstract}{%
  \cleardoublepage
  \thispagestyle{plain}
  \abstract@toptext\par
  \vspace*{8.6mm}
  \begin{center}
    \sloppy\nohyphens{\MakeUppercase\local@title}\par
    \vspace*{13.2mm}
    \@authname\ \@authsurn \par
    \vspace*{7mm}
    \local@monthname/\number\year
  \end{center}\par
  \vspace*{1.5cm}
  \noindent%
  \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
    \local@advisorstring: &
    \count1=0
    \toks@={}
    \@whilenum \count1<\@advisor \do{%
    \ifcase\count1 % same as \ifnum0=\count1
      \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
      \expandafter\endcsname\expandafter\space%
      \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
    \else
      \toks@=\expandafter\expandafter\expandafter{%
        \expandafter\the\expandafter\toks@
        \expandafter&\expandafter\space
        \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
        \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
      }%
    \fi
    \advance\count1 by 1}
    \the\toks@
  \end{tabularx}\par
  \vspace*{2mm}
  \noindent\local@deptstring: \local@deptname\par
  \vspace*{7mm}}{\vspace*{\fill}}
```

`foreignabstract` (*env.*)

```
\newenvironment{foreignabstract}{%
  \cleardoublepage
  \thispagestyle{plain}
  \begin{otherlanguage}{english}
  \foreignabstract@toptext\par
  \vspace*{8.6mm}
  \begin{center}
    \sloppy\nohyphens{\MakeUppercase\foreign@title}\par
    \vspace*{13.2mm}
    \@authname\ \@authsurn \par
    \vspace*{7mm}
    \foreign@monthname/\number\year
  \end{center}\par
  \vspace*{1.5cm}
```

```
  \noindent%
  \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
    \foreign@advisorstring: &
    \count1=0
    \toks@={}
    \@whilenum \count1<\@advisor \do{%
    \ifcase\count1 % same as \ifnum0=\count1
      \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
      \expandafter\endcsname\expandafter\space%
      \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
    \else
      \toks@=\expandafter\expandafter\expandafter{%
        \expandafter\the\expandafter\toks@
        \expandafter&\expandafter\space
        \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
        \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
      }%
    \fi
    \advance\count1 by 1}
    \the\toks@
  \end{tabularx}\par
  \vspace*{2mm}
  \noindent\foreign@deptstring: \foreign@deptname\par
  \vspace*{7mm}}{%
  \end{otherlanguage}
  \vspace*{\fill}
  \global\let\@author\@empty
  \global\let\@date\@empty
  \global\let\foreign@title\@empty
  \global\let\foreign@title\relax
  \global\let\local@title\@empty
  \global\let\local@title\relax
  \global\let\author\relax
  \global\let\author\relax
  \global\let\date\relax}
```

\listoffigures

```
\renewcommand\listoffigures{%
    \coppe@hasLof
    \if@twocolumn
      \@restonecoltrue\onecolumn
    \else
      \@restonecolfalse
    \fi
    \chapter*{\listfigurename}%
      \addcontentsline{toc}{chapter}{\listfigurename}%
      \@mkboth{\MakeUppercase\listfigurename}%
              {\MakeUppercase\listfigurename}%
    \@starttoc{lof}%
    \if@restonecol\twocolumn\fi
    }
```

\listoftables

```
\renewcommand\listoftables{%
```

```
    \if@twocolumn
      \@restonecoltrue\onecolumn
    \else
      \@restonecolfalse
    \fi
    \chapter*{\listtablename}%
      \addcontentsline{toc}{chapter}{\listtablename}%
      \@mkboth{%
          \MakeUppercase\listtablename}%
         {\MakeUppercase\listtablename}%
    \@starttoc{lot}%
    \if@restonecol\twocolumn\fi
    }
```

\printlosymbols

```
\newcommand\printlosymbols{%
\renewcommand\glossaryname{\listsymbolname}%
\@input@{\jobname.los}}
```

\makelosymbols

```
\def\makelosymbols{%
  \newwrite\@losfile
  \immediate\openout\@losfile=\jobname.syx
  \newcommand\symbl[3][]{\@bsphack\begingroup
  \ifstrempty{##1}{\def\@tempsymbl{##2=}}{\def\@tempsymbl{##1=}}%
             \@sanitize%
             \@wrlos{\@tempsymbl}{##2}{##3}}\typeout%
  {Writing index of symbols file \jobname.syx}%
  \let\makelosymbols\@empty%
}%
\@onlypreamble\makelosymbols
```

```
\AtBeginDocument{%
\@ifpackageloaded{hyperref}{%
  \newcommand\@wrlos[3]{%
    \protected@write\@losfile{}%
      {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}%
    \endgroup%
    \@esphack}}{%
  \newcommand\@wrlos[3]{%
    \protected@write\@losfile{}%
      {\string\indexentry{#1[#2] #3}{\thepage}}%
    \endgroup%
    \@esphack}}}%
```

\printloabbreviations

```
\newcommand\printloabbreviations{%
\renewcommand\glossaryname{\listabbreviationname}%
\@input@{\jobname.lab}}
```

\makeloabbreviations

```
\def\makeloabbreviations{%
  \newwrite\@labfile
  \immediate\openout\@labfile=\jobname.abx
```

```
  \newcommand\abbrev[3][]{\@bsphack\begingroup
              \ifstrempty{##1}{\def\@tempsymbl{##2=}}{\def\@tempsymbl{##1=}}
              \@sanitize
              \@wrlab{\@tempsymbl}{##2}{##3}}\typeout
 {Writing index of abbreviations file \jobname.abx}%
 \let\makeloabbreviations\@empty
}
\@onlypreamble\makeloabbreviations

\AtBeginDocument{%
\@ifpackageloaded{hyperref}{%
 \newcommand\@wrlab[3]{%
   \protected@write\@labfile{}%
     {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}%
   \endgroup%
   \@esphack}}{%
 \newcommand\@wrlab[3]{%
   \protected@write\@labfile{}%
     {\string\indexentry{#1[#2] #3}{\thepage}}%
   \endgroup%
   \@esphack}}}%
```

`\newsigla` `\sigla` Acronyms (siglas), manual2024 2.8 / R43. `\newsigla{key}{SIGLA}{full form}` declares one; `\sigla{key}` prints “full form (SIGLA)” on first use and registers it in the list of abbreviations and acronyms (through `\abbrev`, so `\makeloabbreviations` must be active), then just “SIGLA” on later uses. `\sigla*{key}` forces the full form. The first-use flag is set globally so it survives groups (floats, footnotes).

```
\newcommand\newsigla[3]{%
 \@namedef{coppe@sigla@s@#1}{#2}%
 \@namedef{coppe@sigla@l@#1}{#3}}
\newcommand\coppe@sigla@reg[1]{%
 \@ifundefined{abbrev}{}%
   {\abbrev{\@nameuse{coppe@sigla@s@#1}}{\@nameuse{coppe@sigla@l@#1}}}}
\newcommand\coppe@sigla@full[1]{%
 \@nameuse{coppe@sigla@l@#1}\space(\@nameuse{coppe@sigla@s@#1})}
\newcommand\coppe@sigla@undef[1]{%
 \PackageWarning{coppe}{Sigla '#1' nao declarada com \string\newsigla}%
 \textbf{??#1??}}
\newcommand\coppe@sigla@first[1]{%
 \global\@namedef{coppe@sigla@done@#1}{}\coppe@sigla@reg{#1}\coppe@sigla@full{#1}}
\newcommand\coppe@sigla@norm[1]{%
 \@ifundefined{coppe@sigla@s@#1}{\coppe@sigla@undef{#1}}%
   {\@ifundefined{coppe@sigla@done@#1}{\coppe@sigla@first{#1}}%
     {\@nameuse{coppe@sigla@s@#1}}}}
\newcommand\coppe@sigla@star[1]{%
 \@ifundefined{coppe@sigla@s@#1}{\coppe@sigla@undef{#1}}%
   {\global\@namedef{coppe@sigla@done@#1}{}\coppe@sigla@full{#1}}}
\newcommand\sigla{\@ifstar\coppe@sigla@star\coppe@sigla@norm}

%%%% \AtBeginDocument{%
%%%% \@ifpackageloaded{hyperref}{%
%%%%   \def\@wrlab#1#2{%
%%%% \protected@write\@labfile{}%
%%%%       {\string\indexentry{[#1] #2|hyperpage}{\thepage}}%
```

```
%%%%      \endgroup
%%%%      \@esphack}}{%
%%%%    \def\@wrlab#1#2{%
%%%%      \protected@write\@labfile{}%
%%%%        {\string\indexentry{[#1] #2}{\arabic{page}}}%
%%%%      \endgroup
%%%%      \@esphack}}
% Some macros used to generate cataloging information.
\AtBeginDocument{%
  \ltx@ifpackageloaded{hyperref}{
    \def\coppe@bibEnd{%
      \immediate\write\@auxout{%
        \string\newlabel{bib:end}{{}{\arabic{page}}{}{page.\arabic{page}}{}}}}%
    \def\coppe@bibBegin{%
      \immediate\write\@auxout{%
        \string\newlabel{bib:begin}{{}{\arabic{page}}{}{page.\arabic{page}}{}}}}%
    \def\coppe@mainBegin{%
      \immediate\write\@auxout{%
        \string\newlabel{front:pageno}{{}{\Roman{page}}{}{page.\roman{page}}{}}}}%
    \def\coppe@hasLof{%
      \immediate\write\@auxout{%
        \string\newlabel{cat:lofflag}{{}{:~il.;}{}{page.\roman{page}}{}}}}%
  }{%
    \def\coppe@bibEnd{%
      \immediate\write\@auxout{%
      \string\newlabel{bib:end}{{}{\arabic{page}}{}}}}%
    \def\coppe@bibBegin{%
      \immediate\write\@auxout{%
      \string\newlabel{bib:begin}{{}{\arabic{page}}{}}}}%
    \def\coppe@mainBegin{%
      \immediate\write\@auxout{%
        \string\newlabel{front:pageno}{{}{\Roman{page}}{}}}}%
    \def\coppe@hasLof{%
      \immediate\write\@auxout{%
        \string\newlabel{cat:lofflag}{{}{:~il.;}{}}}}%
  }%
}
% The reference list is now produced by biblatex's \printbibliography (see the
% bibliography setup after babel, above). The former thebibliography
% redefinition and its \bibindent are no longer used and have been removed; the
% bib:begin/bib:end catalog labels are now written by \AtBeginBibliography /
% \AtEndBibliography via \coppe@bibBegin and \coppe@bibEnd, defined just above.
```

longquote (*env.*)

```
\newlength{\recuolongquote}%
\setlength{\recuolongquote}{4cm}%
\newenvironment*{longquote}[1][default]{%
 \list{}%
 \footnotesize%
 \addtolength{\leftskip}{\recuolongquote}%
 \item[]%
 \singlespacing%
 \ifthenelse{\not\equal{#1}{default}}{\itshape\selectlanguage{#1}}{}%
}{\endlist}%
```

```
\newenvironment{theglossary}{%
  \if@twocolumn%
    \@restonecoltrue\onecolumn%
  \else%
    \@restonecolfalse%
  \fi%
  \@mkboth{\MakeUppercase\glossaryname}%
  {\MakeUppercase\glossaryname}%
  \chapter*{\glossaryname}%
  \addcontentsline{toc}{chapter}{\glossaryname}
  \list{}
  {\setlength{\listparindent}{0in}%
   \setlength{\labelwidth}{1.0in}%
   \setlength{\leftmargin}{1.5in}%
   \setlength{\labelsep}{0.5in}%
   \setlength{\itemindent}{0in}}%
   \sloppy}%
  {\if@restonecol\twocolumn\fi%
\endlist}
%
\renewenvironment{theindex}{%
  \if@twocolumn
    \@restonecolfalse
  \else
    \@restonecoltrue
  \fi
  \twocolumn[\@makeschapterhead{\indexname}]%
  \@mkboth{\MakeUppercase\indexname}%
  {\MakeUppercase\indexname}%
  \thispagestyle{plain}\parindent\z@
  \addcontentsline{toc}{chapter}{\indexname}
  \parskip\z@ \@plus .3\p@\relax
  \columnseprule \z@
  \columnsep 35\p@
  \let\item\@idxitem}
  {\if@restonecol\onecolumn\else\clearpage\fi}
\if@english
  \newcommand\listabbreviationname{List of Abbreviations and Acronyms}
  \newcommand\listsymbolname{List of Symbols}
  \newcommand\glossaryname{Glossary}
\else
  \newcommand\listabbreviationname{Lista de Abreviaturas e Siglas}
  \newcommand\listsymbolname{Lista de S{\' i}mbolos}
  \newcommand\glossaryname{Gloss{\' a}rio}
\fi
%
\newcommand\local@advisorstring{Orientador}
\newcommand\foreign@advisorstring{Advisor}
\ifthenelse{\boolean{maledoc}}{%
  \newcommand\local@approvedname{Examinado por}%
}{%
  \newcommand\local@approvedname{Examinada por}%
}
\newcommand\local@universityname{Universidade Federal do Rio de Janeiro}
```

```
\newcommand\local@deptstring{Programa}
\newcommand\foreign@deptstring{Department}
\newcommand\local@cityname{Rio de Janeiro}
\newcommand\local@statename{RJ}
\newcommand\local@countryname{Brasil}
%
\newcommand\frontcover@maintext{
  \sloppy\nohyphens{\local@doctype\ de \@degreename\
  \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
  ao Programa de P{\' o}s-gradua{\c c}{\~ a}o em \local@deptname,
  COPPE, da \local@universityname, como parte dos requisitos
  necess{\' a}rios {\` a} obten{\c c}{\~ a}o do t{\' i}tulo de
  \local@degname\ em \local@deptname.}
}
%
\newcommand\frontpage@maintext{
  \noindent {\MakeUppercase\local@doctype}
  \ifthenelse{\boolean{maledoc}}{SUBMETIDO}{SUBMETIDA}
  \sloppy\nohyphens{AO CORPO DOCENTE DO INSTITUTO ALBERTO LUIZ COIMBRA
  DE P{\' O}S-GRADUA{\c C}{\~ A}O E PESQUISA DE ENGENHARIA DA
  UNIVERSIDADE FEDERAL DO RIO DE JANEIRO COMO PARTE DOS REQUISITOS
  NECESS{\' A}RIOS PARA A OBTEN{\c C}{\~ A}O DO GRAU DE
  {\MakeUppercase\local@degname} EM CI{\^E}NCIAS EM
  {\MakeUppercase\local@deptname.\par}}%
}
%
\newcommand\frontpage@bottomtext{%
  \begin{center}
  {\MakeUppercase{\local@cityname, \local@statename\ -- \local@countryname}}\par
  {\MakeUppercase\local@monthname\ DE \number\year}
  \end{center}%
}
%
\newcommand\abstract@toptext{%
  \noindent Resumo \ifthenelse{\boolean{maledoc}}{do}{da}
  \local@doctype\ \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
  \sloppy\nohyphens{{\` a} COPPE/UFRJ como parte dos requisitos
  necess{\' a}rios para a obten{\c c}{\~ a}o do grau de
  \local@degname\ em Ci{\^ e}ncias (\@degree)}
}
\newcommand\foreignabstract@toptext{%
  \noindent \sloppy\nohyphens{Abstract of \foreign@doctype\ presented to
  COPPE/UFRJ as a partial fulfillment of the requirements for the
  degree of \foreign@degname\ of Science (\@degree)}
}
%
```

⟨/class⟩

⟨*glossary⟩

```
actual '='
quote '!'
level '>'
%%%% delim_0   ", p. "
delim_0 "\\dotfill "
```

```
lethead_flag  0
headings_flag 0
preamble
"\n\\begin{theglossary}\n  \\makeatletter"
postamble
"\n  \\end{theglossary}\n"
⟨/glossary⟩
```

```
⟨*class⟩
\newcommand{\annex}{
    \iflanguage{brazilian}
        {\renewcommand{\appendixname}{Anexo}}
        {\renewcommand{\appendixname}{Annex}}
    \appendix
}
⟨/class⟩
```

## BibLaTeX style files

Generated from this .dtx by coppe.ins (one docstrip module each).

```
⟨*dbx⟩
\ProvidesFile{coppe.dbx}[2026/05/23 v0.2 CoppeTeX ABNT datamodel]
% Custom entry types beyond biblatex's standard datamodel (games, TV, maps).
% map = cartographic document (NBR 6023 4.2.12); biblatex has no native @map.
% The remaining ABNT "exotic" types (legislation, jurisdiction, legal, patent,
% standard, music, audio, image, artwork, video, dataset, letter, periodical,
% performance) are already in biblatex's standard datamodel.
\DeclareDatamodelEntrytypes{game,videogame,boardgame,tvshow,map}
% Custom fields.
\DeclareDatamodelFields[type=field,datatype=literal]{course}      % "(... em <course>)"
\DeclareDatamodelFields[type=field,datatype=literal]{coppedegree} % degree word
\DeclareDatamodelFields[type=field,datatype=literal]{platform}    % videogame platform
\DeclareDatamodelFields[type=field,datatype=literal]{scale}       % map scale (escala)
\DeclareDatamodelFields[type=list,datatype=literal]{artist}       % game/illustration art
\DeclareDatamodelFields[type=list,datatype=literal]{director}     % movie/tvshow/video
\DeclareDatamodelFields[type=list,datatype=literal]{producer}
\DeclareDatamodelEntryfields{course,coppedegree}
\DeclareDatamodelEntryfields[game,videogame,boardgame,software,movie,tvshow,video]%
  {artist,platform,director,producer,version}
\DeclareDatamodelEntryfields[map]{scale}
⟨/dbx⟩
⟨*bbx⟩
\ProvidesFile{coppe.bbx}[2026/05/23 v0.1 CoppeTeX ABNT bibliography style]
% ABNT reference list keeps the date at the end (imprenta), same for author-date
% and numeric modes, so we derive from the standard style, not authoryear.
\RequireBibliographyStyle{standard}

% Note: standard style never dashes repeated authors, which already matches the
% post-2020 ABNT rule (repeat the entry in full), so no 'dashed' option is needed.
\ExecuteBibliographyOptions{%
  maxbibnames=99,      % ABNT 4.3.1.3: list all authors by default
  giveninits=false,
  uniquename=false,
  uniquelist=false,
```

```
  labeldateparts=true, % needed for author-date extradate (1999a/1999b)
  sorting=nyt,
}

% Language strings (Disponível em / Acesso em / In, etc.)
\DeclareLanguageMapping{brazilian}{brazilian-coppe}
\DeclareLanguageMapping{english}{english-coppe}

% --- Names: all reversed "FAMILY, Given", separated by ";" in the list ---
\DeclareNameAlias{author}{family-given}
\DeclareNameAlias{editor}{family-given}
\DeclareNameAlias{translator}{family-given}
\DeclareNameAlias{default}{family-given}
\DeclareDelimFormat[bib,biblist]{multinamedelim}{\addsemicolon\space}
\DeclareDelimFormat[bib,biblist]{finalnamedelim}{\addsemicolon\space}

% Uppercase the surname (family + prefix) only inside the bibliography list;
% citations keep normal caps (post-2020 rule).
\AtBeginBibliography{%
  \singlespacing                         % entries single-spaced internally (2.4)
  \setlength{\bibitemsep}{\baselineskip}% one blank line between entries (2.4)
  \renewcommand*{\mkbibnamefamily}[1]{\MakeUppercase{#1}}%
  \renewcommand*{\mkbibnameprefix}[1]{\MakeUppercase{#1}}%
}

% --- Titles: bold for whole-document types; plain for parts; entry-by-title
%     (no author/editor) -> not bold, first significant word in CAPS (ABNT 4.2). ---
% entry-by-title: plain (not bold). TODO: first significant word in CAPS (ABNT 4.2)
% once a robustly-tested uppercase-first-word macro is in place.
\DeclareFieldFormat{title}{%
  \ifboolexpr{ test {\ifnameundef{labelname}} }
    {#1}%
    {\mkbibbold{#1}}}
\DeclareFieldFormat[article,incollection,inbook,inproceedings,suppperiodical,review]{title}{
\DeclareFieldFormat[thesis]{title}{\mkbibbold{#1}}
\DeclareFieldFormat{journaltitle}{\mkbibbold{#1}}
\DeclareFieldFormat{booktitle}{\mkbibbold{#1}}
\DeclareFieldFormat{maintitle}{\mkbibbold{#1}}

% --- Pages: ABNT uses "p." (also for ranges); theses use "f." for pagetotal ---
\DeclareFieldFormat{pagetotal}{#1\addnbspace p\adddot}
\DeclareFieldFormat[thesis]{pagetotal}{#1\addnbspace f\adddot}
\DeclareFieldFormat{pages}{p\adddot\addnbspace#1}

% --- Periodical volume / number prefixes ---
\DeclareFieldFormat[article]{volume}{v\adddot\addnbspace#1}
\DeclareFieldFormat[article]{number}{n\adddot\addnbspace#1}

% --- "In:" before host title (incollection/inproceedings/inbook) ---
\renewbibmacro*{in:}{\printtext{\bibstring{in}\addcolon\space}}

% --- Online: "Disponível em: <url>." + "Acesso em: <date>." via lbx strings ---
\DeclareFieldFormat{url}{\bibstring{availablefrom}\addcolon\space\url{#1}}

```

```
% --- ABNT article driver: no "In:"; Journal, Local, v., n., p., date, DOI/URL ---
\DeclareBibliographyDriver{article}{%
  \usebibmacro{bibindex}%
  \usebibmacro{begentry}%
  \usebibmacro{author/translator+others}%
  \setunit{\labelnamepunct}\newblock
  \usebibmacro{title}%
  \newunit\newblock
  \printfield{journaltitle}%
  \setunit{\addcomma\space}\printlist{location}%
  \setunit{\addcomma\space}\printfield{volume}%
  \setunit{\addcomma\space}\printfield{number}%
  \setunit{\addcomma\space}\printfield{pages}%
  \setunit{\addcomma\space}\printdate%
  \newunit\newblock
  \printfield{doi}%
  \newunit\newblock
  \usebibmacro{url+urldate}%
  \usebibmacro{finentry}}

% --- type field: resolve a known bibliography string (babel-aware), else literal ---
\DeclareFieldFormat{type}{\ifbibstring{#1}{\bibstring{#1}}{#1}}

% --- Thesis: "<type> (em <course>)" using the custom course field (ABNT). ---
\DeclareFieldFormat{course}{\mkbibparens{\iflanguage{brazilian}{em\addspace}{in\addspace}#1}
\DeclareBibliographyDriver{thesis}{%
  \usebibmacro{bibindex}\usebibmacro{begentry}%
  \usebibmacro{author}%
  \setunit{\labelnamepunct}\newblock
  \usebibmacro{title}%
  \newunit\newblock
  \printfield{type}\setunit*{\addspace}\printfield{course}%
  \newunit\newblock
  \printlist{institution}%
  \setunit*{\addcomma\space}\printlist{location}%
  \setunit*{\addcomma\space}\printdate%
  \newunit\newblock
  \printfield{pagetotal}%
  \newunit\newblock
  \printfield{note}%
  \newunit\newblock
  \usebibmacro{url+urldate}%
  \usebibmacro{finentry}}
% --- Games / audiovisual: render the custom fields (artist, platform,
%     director, producer, version) the generic misc driver ignores. ---
\newbibmacro*{coppe:media}{%
  \iflistundef{artist}{}{\setunit{\addperiod\space}%
    \printtext{\iflanguage{brazilian}{Ilustra\c{c}\~ao}{Art}\addcolon\space}\printlist{artis
  \iffieldundef{platform}{}{\setunit{\addperiod\space}%
    \printtext{\iflanguage{brazilian}{Plataforma}{Platform}\addcolon\space}\printfield{platf
  \iflistundef{director}{}{\setunit{\addperiod\space}%
    \printtext{\iflanguage{brazilian}{Dire\c{c}\~ao}{Directed by}\addcolon\space}\printlist{
  \iflistundef{producer}{}{\setunit{\addperiod\space}%
    \printtext{\iflanguage{brazilian}{Produ\c{c}\~ao}{Produced by}\addcolon\space}\printlist
```

```
\newbibmacro*{coppe:mediadriver}{%
  \usebibmacro{bibindex}\usebibmacro{begentry}%
  \usebibmacro{author/translator+others}%
  \setunit{\labelnamepunct}\newblock
  \usebibmacro{title}%
  \newunit\newblock
  \usebibmacro{coppe:media}%
  \newunit\newblock
  \printlist{publisher}%
  \setunit*{\addcomma\space}\printlist{location}%
  \setunit*{\addcomma\space}\printdate%
  \newunit\newblock
  \printfield{version}%
  \newunit\newblock
  \printfield{note}%
  \newunit\newblock
  \usebibmacro{url+urldate}%
  \usebibmacro{finentry}}
\DeclareBibliographyDriver{game}{\usebibmacro{coppe:mediadriver}}
\DeclareBibliographyDriver{videogame}{\usebibmacro{coppe:mediadriver}}
\DeclareBibliographyDriver{boardgame}{\usebibmacro{coppe:mediadriver}}
\DeclareBibliographyDriver{movie}{\usebibmacro{coppe:mediadriver}}
\DeclareBibliographyDriver{tvshow}{\usebibmacro{coppe:mediadriver}}
% Cartographic document (map): rendered like a book (author/title/imprenta).
\DeclareBibliographyDriver{map}{\usedriver{}{book}}
% Exotic ABNT types that biblatex's standard style leaves without a driver:
% give each a baseline (book-like for published docs: norma/partitura; misc for
% the rest) so they print. Per-type ABNT refinements can follow later.
\DeclareBibliographyDriver{standard}{\usedriver{}{book}}
\DeclareBibliographyDriver{music}{\usedriver{}{book}}
\DeclareBibliographyDriver{legislation}{\usedriver{}{misc}}
\DeclareBibliographyDriver{jurisdiction}{\usedriver{}{misc}}
\DeclareBibliographyDriver{legal}{\usedriver{}{misc}}
\DeclareBibliographyDriver{audio}{\usedriver{}{misc}}
\DeclareBibliographyDriver{video}{\usedriver{}{misc}}
\DeclareBibliographyDriver{software}{\usedriver{}{misc}}
\DeclareBibliographyDriver{image}{\usedriver{}{misc}}
\DeclareBibliographyDriver{artwork}{\usedriver{}{misc}}
\DeclareBibliographyDriver{performance}{\usedriver{}{misc}}
\DeclareBibliographyDriver{dataset}{\usedriver{}{misc}}

% --- biber source map: Portuguese synonyms (entry types + fields) + thesis family ---
\DeclareSourcemap{%
  \maps[datatype=bibtex]{%
    % entry-type PT synonyms -> canonical English
    \map{\step[typesource=livro,typetarget=book]}
    \map{\step[typesource=artigo,typetarget=article]}
    \map{\step[typesource=artigodejornal,typetarget=article]}
    \map{\step[typesource=partedelivro,typetarget=incollection]}
    \map{\step[typesource=emlivro,typetarget=inbook]}
    \map{\step[typesource=verbete,typetarget=inreference]}
    \map{\step[typesource=anais,typetarget=proceedings]}
    \map{\step[typesource=trabalhodeevento,typetarget=inproceedings]}
    \map{\step[typesource=periodico,typetarget=periodical]}
```

```
\map{\step[typesource=relatorio,typetarget=report]}
\map{\step[typesource=relatoriotecnico,typetarget=report]}
\map{\step[typesource=manual,typetarget=manual]}
\map{\step[typesource=correspondencia,typetarget=letter]}
\map{\step[typesource=patente,typetarget=patent]}
\map{\step[typesource=diverso,typetarget=misc]}
\map{\step[typesource=jogo,typetarget=game]}
\map{\step[typesource=jogoeletronico,typetarget=videogame]}
\map{\step[typesource=jogodetabuleiro,typetarget=boardgame]}
\map{\step[typesource=filme,typetarget=movie]}
\map{\step[typesource=serietv,typetarget=tvshow]}
\map{\step[typesource=programadetv,typetarget=tvshow]}
% Exotic ABNT types (NBR 6023 4.2.5-4.2.14); targets are biblatex-native
% except map (declared in coppe.dbx).
\map{\step[typesource=legislacao,typetarget=legislation]}
\map{\step[typesource=atonormativo,typetarget=legislation]}
\map{\step[typesource=jurisprudencia,typetarget=jurisdiction]}
\map{\step[typesource=documentojuridico,typetarget=legal]}
\map{\step[typesource=documentocivil,typetarget=legal]}
\map{\step[typesource=norma,typetarget=standard]}
\map{\step[typesource=partitura,typetarget=music]}
\map{\step[typesource=iconografico,typetarget=image]}
\map{\step[typesource=obradearte,typetarget=artwork]}
\map{\step[typesource=tridimensional,typetarget=artwork]}
\map{\step[typesource=cartografico,typetarget=map]}
\map{\step[typesource=mapa,typetarget=map]}
\map{\step[typesource=audiovisual,typetarget=video]}
\map{\step[typesource=conjuntodedados,typetarget=dataset]}
\map{\step[typesource=fasciculo,typetarget=suppperiodical]}
\map{\step[typesource=trabalhoapresentado,typetarget=unpublished]}
% thesis family: retype to thesis and preset the (babel-aware) type word
\map{\step[typesource=tese,typetarget=thesis]}
\map{\step[typesource=dissertacao,typetarget=thesis]}
\map{\step[typesource=masterdissertation,typetarget=thesis,final]\step[fieldset=type,fie
\map{\step[typesource=mscdissertation,typetarget=thesis,final]\step[fieldset=type,fieldv
\map{\step[typesource=dissertacaomestrado,typetarget=thesis,final]\step[fieldset=type,fi
\map{\step[typesource=dscthesis,typetarget=thesis,final]\step[fieldset=type,fieldvalue=d
\map{\step[typesource=tesedoutorado,typetarget=thesis,final]\step[fieldset=type,fieldval
\map{\step[typesource=phdthesis,typetarget=thesis,final]\step[fieldset=type,fieldvalue=p
\map{\step[typesource=tesephd,typetarget=thesis,final]\step[fieldset=type,fieldvalue=phd
\map{\step[typesource=tcc,typetarget=thesis,final]\step[fieldset=type,fieldvalue=tcc]}
% field PT synonyms (unaccented) -> canonical English
\map{\step[fieldsource=titulo,fieldtarget=title]}
\map{\step[fieldsource=subtitulo,fieldtarget=subtitle]}
\map{\step[fieldsource=autor,fieldtarget=author]}
\map{\step[fieldsource=editora,fieldtarget=publisher]}
\map{\step[fieldsource=local,fieldtarget=location]}
\map{\step[fieldsource=ano,fieldtarget=year]}
\map{\step[fieldsource=data,fieldtarget=date]}
\map{\step[fieldsource=edicao,fieldtarget=edition]}
\map{\step[fieldsource=paginas,fieldtarget=pages]}
\map{\step[fieldsource=numero,fieldtarget=number]}
\map{\step[fieldsource=tradutor,fieldtarget=translator]}
\map{\step[fieldsource=organizador,fieldtarget=editor]}
```

```
    \map{\step[fieldsource=instituicao,fieldtarget=institution]}
    \map{\step[fieldsource=tipo,fieldtarget=type]}
    \map{\step[fieldsource=nota,fieldtarget=note]}
    \map{\step[fieldsource=dataacesso,fieldtarget=urldate]}
    \map{\step[fieldsource=curso,fieldtarget=course]}
    \map{\step[fieldsource=grau,fieldtarget=coppedegree]}
    \map{\step[fieldsource=artista,fieldtarget=artist]}
    \map{\step[fieldsource=plataforma,fieldtarget=platform]}
    \map{\step[fieldsource=diretor,fieldtarget=director]}
    \map{\step[fieldsource=produtor,fieldtarget=producer]}
    \map{\step[fieldsource=versao,fieldtarget=version]}
    \map{\step[fieldsource=escala,fieldtarget=scale]}
  }%
}

% --- ABNT list layout: flush left, single spacing, blank line between entries ---
\setlength{\bibhang}{0pt}
\setlength{\bibitemsep}{\baselineskip}

⟨/bbx⟩
⟨*cbx⟩
\ProvidesFile{coppe.cbx}[2026/05/23 v0.1 CoppeTeX ABNT author-date citations]
% Author-date is the CoppeTeX 4.0 default. authoryear-comp gives compressed
% author-year cites with normal-caps names (post-2020 rule) out of the box.
\RequireCitationStyle{authoryear-comp}

% --- NBR 10520:2023 author separator -----------------------------------------
% Parenthetical citations separate authors with ";" -> (Silva; Souza, 2020).
% Narrative citations (\textcite, natbib \citet) keep comma + "e/and" ->
% Silva, Souza e Oliveira (2020). The reference list keeps its own ";" via
% coppe.bbx ([bib,biblist]); these contextless formats cover the cite side.
\DeclareDelimFormat{multinamedelim}{\addsemicolon\space}
\DeclareDelimFormat{finalnamedelim}{\addsemicolon\space}
\DeclareDelimFormat[textcite]{multinamedelim}{\addcomma\space}
\DeclareDelimFormat[textcite]{finalnamedelim}{\addspace\bibstring{and}\space}

% --- apud: citation of a citation (ABNT NBR 6023 §4.1.2.2.1) ----------------
% The reference list entry is the CITING work (the one actually consulted).
%   \citetapud[<page>]{original}{citing} -> Original (yr) apud Citing (yr[, page])
%   \citepapud[<page>]{original}{citing} -> (Original, yr apud Citing, yr[, page])
% #1 = optional postnote (page/loc) on the citing work; #2 = original (not
% accessed); #3 = work consulted. "apud" is a latin expression -> italic (§4.1.2.2).
\DeclareRobustCommand{\citetapud}[3][]{%
  \citeauthor{#2}\space(\citeyear{#2})\space\mkbibemph{apud}\space\textcite[#1]{#3}}
\DeclareRobustCommand{\citepapud}[3][]{%
  \mkbibparens{%
    \citeauthor{#2}\addcomma\space\citeyear{#2}\space
    \mkbibemph{apud}\space
    \citeauthor{#3}\addcomma\space\citeyear{#3}%
    \ifblank{#1}{}{\addcomma\space\mkpageprefix[pagination]{#1}}}}
⟨/cbx⟩
⟨*lbxbr⟩
\ProvidesFile{brazilian-coppe.lbx}[2026/05/23 v0.1 CoppeTeX brazilian strings]
\InheritBibliographyExtras{brazilian}
```

```
\NewBibliographyString{availablefrom,mscdiss,dscthesis,tcc}
\DeclareBibliographyStrings{%
  inherit       = {brazilian},
  availablefrom = {{Dispon\'{\i}vel em}{Dispon\'{\i}vel em}},
  urlseen       = {{Acesso em}{Acesso em}},
  in            = {{In}{In}},
  mscdiss       = {{Disserta\c{c}\~{a}o de Mestrado}{Disserta\c{c}\~{a}o de Mestrado}},
  dscthesis     = {{Tese de Doutorado}{Tese de Doutorado}},
  phdthesis     = {{Tese de Doutorado}{Tese de Doutorado}},
  tcc           = {{Trabalho de Conclus\~{a}o de Curso}{Trabalho de Conclus\~{a}o de Curso}}
}
⟨/lbxbr⟩
⟨*lbxen⟩
\ProvidesFile{english-coppe.lbx}[2026/05/23 v0.1 CoppeTeX english strings]
\InheritBibliographyExtras{english}
\NewBibliographyString{availablefrom,mscdiss,dscthesis,tcc}
\DeclareBibliographyStrings{%
  inherit       = {english},
  availablefrom = {{Available from}{Available from}},
  urlseen       = {{Access on}{Access on}},
  in            = {{In}{In}},
  mscdiss       = {{M\adddot Sc\adddot\ Dissertation}{M\adddot Sc\adddot\ Diss\adddot}},
  dscthesis     = {{D\adddot Sc\adddot\ Thesis}{D\adddot Sc\adddot\ Thesis}},
  phdthesis     = {{Ph\adddot D\adddot\ Thesis}{Ph\adddot D\adddot\ Thesis}},
  tcc           = {{Undergraduate Thesis}{Undergraduate Thesis}},
}
⟨/lbxen⟩
⟨*numbbx⟩
\ProvidesFile{coppe-numeric.bbx}[2026/05/24 v0.1 CoppeTeX ABNT numeric bibliography]
% Numeric reference list for the ABNT numeric system (NBR 10520). It reuses the
% full coppe ABNT style (drivers, CAPS surnames, bold titles, "In:", p./f.,
% the PT->EN source map and the language mappings) and only adds the [n] list
% label and the numeric sorting on top.
\RequireBibliographyStyle{coppe}

% ABNT numeric system lists references in the order they are first cited
% (NBR 10520), i.e. unsorted. Override with sorting=nty for an alphabetical
% numbered list. 'labelnumber' generates the [n] field.
\ExecuteBibliographyOptions{labelnumber,sorting=none}

% Bracketed labels, both in the list and for any shorthand entries.
\DeclareFieldFormat{labelnumberwidth}{\mkbibbrackets{#1}}
\DeclareFieldFormat{shorthandwidth}{\mkbibbrackets{#1}}

% Reference list driven by the [n] label (copied from biblatex's numeric.bbx),
% keeping the ABNT blank line between entries via \bibitemsep set in coppe.bbx.
\defbibenvironment{bibliography}
  {\list
     {\printtext[labelnumberwidth]{%
        \printfield{labelprefix}%
        \printfield{labelnumber}}}
     {\setlength{\labelwidth}{\labelnumberwidth}%
      \setlength{\leftmargin}{\labelwidth}%
      \setlength{\labelsep}{\biblabelsep}%
```

```
      \addtolength{\leftmargin}{\labelsep}%
      \setlength{\itemsep}{\bibitemsep}%
      \setlength{\parsep}{\bibparsep}}%
      \renewcommand*{\makelabel}[1]{\hss##1}}
  {\endlist}
  {\item}

⟨/numbbx⟩
⟨*numcbx⟩
\ProvidesFile{coppe-numeric.cbx}[2026/05/24 v0.1 CoppeTeX ABNT numeric citations]
% Numeric citation system (ABNT NBR 10520): in-text references are bracketed
% numbers [1], compressed and sorted within a single cite group. This is the
% opt-in alternative to the author-date default (coppe.cbx); it pairs with
% bibstyle=coppe-numeric. With the biblatex natbib option, \citep -> [1] and
% \citet -> Author [1].
\RequireCitationStyle{numeric-comp}
⟨/numcbx⟩
```

## Acknowledgments

Thanks to all CoppeTeX users who have reported their experience with this class. We also thank to professor Fernando Lizarralde and Heiko Oberdiek for their helpful comments. The authors would like to thank the National Council for Scientific and Technological Development (CNPq) of Brazil.

## Index

Numbers written in italic refer to the page where the corresponding entry is described; numbers underlined refer to the code line of the definition; numbers in roman refer to the code lines where the entry is used.

# Change History